

HORTICULTURAL CROP PRODUCTION (AGT 226)

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OBJECTIVES

By the end of this unit, you will be able to:

- Understand the scope of horticulture and classify horticultural crops.
- Understand the different methods of propagating horticultural plants.
- Understand the principles of soil sterilization for horticultural crop nurseries
- Understand the principles and techniques of fruit crop cultivation, orchard establishment and maintenance.
- Understand the principles and practice of cultivating various types of vegetables
- Understand the principles and practices of ornamental horticulture.

HORTICULTURE is a specialized branch of agriculture focused on the intensive production of garden crops, namely fruits, vegetables, and ornamental plants. Derived from the Latin words hortus (garden) and colere (to cultivate), it bridges small-scale gardening and large-scale field farming.

DEFINITION OF HORTICULTURE

The word horticulture is derived from two latin words “Hortus” meaning garden and “Culture” meaning the cultivate. Horticulture is therefore concerned with the cultivation of garden plants such as fruits, Vegetables, flowers and ornamental plants. Horticulture may be defined as the science and art of growing fruits, vegetables, flowers, plantation crops and ornamental crops for man, industrial and aesthetic uses.

When horticulture is considered from both the science and business perspectives, horticulture can be broadly defined as the science and art of cultivating, processing and marketing of fruits, vegetables, flowers and ornamental plants.

Horticulture is a branch of agriculture.

Horticulture is subdivided into five; namely pomology, olericulture, floriculture, arboriculture and landscaping.

Pomology: - The science and practice of fruit production.

Olericulture: - The science and practice of growing vegetables.

Floriculture: - The science and practice of production of flowers and ornamentals.

Arboriculture: - The science and practice of growing and caring for ornamental trees.
Landscaping: - The science and practice of using plants outdoor for aesthetic and functional purposes.

HISTORY OF HORTICULTURE

Horticulture is as old as mankind since man cannot survive without food. The concepts of garden culture can be traced as far as 3, 000 BC or beyond in Egypt. Historical evidence shows that by this time crops such as grapes, dates, olives and onions have been brought under cultivation by Egyptians and the technology such as land preparation, pruning, irrigation drying etc had also been used. The hanging garden of Babylon which was of flowers and fruits was hailed as one of the seven wonders of the ancient world. During the Middle Ages (500 – 1500 CA), horticultural plants were grown in protected areas surrounded by high walls or similar structures in temperate Europe. As society evolved, deliberate cultivation and domestication of edible plants replaced the less efficient food gathering habits of primitive societies. Agriculture, particularly horticulture, is therefore not a modern day invention but one that continues to be transformed as society advances technologically. The idea of intensive management of horticultural crops still holds today even though some of the vegetables like tomato are grown in the field like other field crops. At present, the fields of horticultural crops are not only grown within the home gardens but also in large quantities as commercial enterprises. Many researches have been conducted aimed at solving the problems of growing horticultural crops and a comparatively large volume of scientific information on behavior of horticultural plants and products have been made available.

IMPORTANCE OF VEGETABLES AND FRUITS IN NIGERIA

The importance of vegetables and fruits in Nigerian economy cannot be overemphasized. Just as crops and animals play a vital role in the economy of Nigeria, so do vegetables and fruits play vital roles in individuals, community and the country at large.

Vegetables and fruits provide the following

Provision of Food – Vegetables and fruits are sources of food which provide carbohydrates, vitamins, minerals and water to humans and animals. Plant source of food are usually regarded as good and natural source of food for human and animals.

Provision of Income – Provision of income per unit area yield of horticultural crops is very high as compared to field crops. This high yield per unit area invariably leads to high returns per unit area which means that the income of the farmer is significantly increased. This therefore, helps reduce poverty. Source of Raw Material – Vegetables and fruits provide raw materials for industries such as fruit juice industries, pharmaceutical industries as well as oil mills. This helps to reduce importation of the raw materials and hence reduce cost of production.

Provision of Employment – Those that are involved in the production, processing and marketing of vegetables and fruits are usually self employed or employed by industries that use these products as raw materials. This reduces idleness and level of unemployment in the country. Uses Waste and Undulating Lands – Fruits and Vegetables can be grown in land where the gradient is uneven or where the land is undulating. Mango, cashew, Vegetables can be grown on large scale on hilly area.

Source of Foreign Exchange – When Vegetables and fruits are exported to other countries this serves as foreign exchange for the country and this increases the country's economy.

Importance of Vegetables in Human Diet

Vegetables supply most of the nutrients that are deficient in other food materials. This includes supply of minerals, especially calcium and iron.

Vegetables are acid neutralizers e.g. Okra, *Corchorus* Spp neutralises the acid produced from the some fruits.

Vegetables prevent constipation and promote digestion as a result of fibres/roughages obtained from Okra, Cucumber, Amaranthus, Lettuce and Cabbage.

Vegetables are rich sources of vitamins A, B, and C which helps to lower susceptibility to infection. e.g.: Carrots, Sweet Corn, Amaranthus And Celosia provide Vitamin A; Bitter leaf, Water leaf, Solanum and Celosia provide Vitamin B; Tomatoes, Carrots, Lettuce, Cabbage and Amaranthus provide Vitamin C.

Some vegetables are rich sources of carbohydrate e.g. Potatoes, Sweet Corn, Carrot etc. Green beans and peas are cheap sources of protein.

Vernonia (Bitter leaf), *Amaranthus* and *Tellfeira* provide some amount of protein in human diet.

Vegetables are generally needed to have balanced diets and overcome nutritional deficiencies.

Vegetables make our staple food more palatable and enhance their intake.

CHARACTERISTICS

1. High Moisture Content and Perishability

Characteristic: Horticultural crops generally possess very high water content, often ranging from 70% to 95% at maturity.

Implication: This makes them highly perishable. Unlike cereal grains, which can be stored dry for years, horticultural crops require immediate consumption, specialized post-harvest handling (like cold chains), or processing (canning, freezing, drying) to prevent rapid degradation by microbes and enzymatic breakdown.

2. Intensive Cultivation and Management

Characteristic: Horticultural plants require intense, specialized care throughout their lifecycle.

Implication: They demand higher inputs of labor, capital, and technology per unit area compared to field crops. Practices include precise irrigation, pruning, training, staking, specialized fertilization, and controlled environment agriculture (greenhouses, hydroponics).

3. Utilization in a Fresh or Living State

Characteristic: A primary distinguishing factor is how these plants are utilized.

Implication: Vegetables and fruits are frequently consumed fresh, requiring strict food safety and cosmetic standards. Ornamental plants and turf grasses are utilized in their living state for aesthetic, functional, and psychological benefits.

4. High Value per Unit Area

Characteristic: Horticultural crops are typically "high-value" crops.

Implication: Because of the intensive management and consumer demand for fresh, high-quality produce and ornamentals, the financial return per acre or hectare is substantially higher than that of staple agronomic crops. This allows them to be profitably grown on smaller plots of land.

5. High Aesthetic Value

Characteristic: Beauty is a primary product of many horticultural plants.

Implication: Unlike agronomy, which focuses almost entirely on utility and yield, floriculture and landscape horticulture focus heavily on visual appeal, fragrance, and structure. Plant breeding in this sector often prioritizes color, shape, and shelf-life over sheer caloric yield.

6. Diverse Propagation Methods

Characteristic: While agronomic crops are almost exclusively grown from seed, horticultural plants heavily utilize vegetative (asexual) propagation.

Implication: Techniques such as grafting, budding, cuttings, and tissue culture are standard. This ensures clonal uniformity, preserves unique genetic traits (like seedless fruits), and allows for the combination of desirable rootstocks and scions (e.g., in apple or citrus production).

Comparative Analysis: Horticulture vs. Agronomy

To better understand horticultural plants, it is helpful to contrast them directly with agronomic (field) crops.

Feature Horticultural Plants Agronomic Plants
Examples Tomatoes, Apples, Roses, Turfgrass Wheat, Maize, Rice, Soybeans

Moisture Content High (Highly perishable) Low (Easily stored dry)
Scale of Production Typically smaller, intensive plots Large-scale, extensive acreage
Management Intensity High (Pruning, hand-harvesting, etc.) Low to Moderate (Highly mechanized)
Economic Value High return per unit area Lower return per unit area
Primary State of Use Often fresh, living, or minimally processed Processed, dried, or milled
Aesthetic Focus High (especially in ornamentals) Low (focus is on yield and utility)

CLASSIFICATION OF HORTICULTURAL PLANTS

Horticultural plants are primarily classified based on their use and the specific branch of horticulture under which they are cultivated. Below are the major types of horticultural plants categorized by discipline.

1. **Olericulture** (Vegetable Crops)

Olericulture is the cultivation of herbaceous plants for human consumption. Vegetables are typically classified by the edible part of the plant.

Fruit Vegetables: Cultivated for their botanical fruits. Examples: Tomatoes, peppers, okra, cucumbers, eggplants.

Leafy Vegetables: Cultivated for their foliage. Examples: Spinach, lettuce, cabbage, amaranth, fluted pumpkin (*Telfairia occidentalis*).

Root and Tuber Vegetables: Cultivated for their swollen underground roots or stems. Examples: Carrots, sweet potatoes, yams, radishes.

Bulb Vegetables: Cultivated for their fleshy, layered underground structures. Examples: Onions, garlic, leeks.

Stem Vegetables: Cultivated for their edible above-ground stems. Examples: Asparagus, kohlrabi, celery.

2. **Pomology** (Fruit and Nut Crops)

Pomology focuses on the cultivation, production, and science of fruit-bearing trees and woody plants.

Tree Fruits: Large woody perennials that produce fleshy fruits. Examples: Mangoes, citrus (oranges, lemons), apples, avocados, guavas.

Small Fruits (Berries & Vines): Smaller plants, vines, or trailing shrubs producing edible fruits. Examples: Grapes, strawberries, pineapples, raspberries, blackberries.

Nut Crops: Woody plants cultivated for their seeds, which are typically encased in a hard shell. Examples: Cashews, walnuts, almonds, pecans, kola nuts.

3. Floriculture (Floral Crops)

Floriculture is the cultivation of flowering and ornamental plants for gardens, floristry, and the floral industry.

Cut Flowers: Flowers harvested with their stems intact for use in arrangements and bouquets.

Examples: Roses, lilies, carnations, chrysanthemums.

Potted Plants: Plants grown and sold primarily in containers for indoor or patio decoration.

Examples: Orchids, poinsettias, ferns, peace lilies.

Bedding Plants: Temporary plantings (often annuals) used to create colorful garden displays.

Examples: Petunias, marigolds, zinnias, begonias.

4. Landscape and Ornamental Horticulture

This branch deals with the cultivation of plants for aesthetic enhancement, functional landscaping, and environmental engineering.

Shade and Avenue Trees: Large trees planted for cooling, aesthetics, or property boundaries.

Examples: Neem trees, flamboyant trees, terminalia, oaks.

Ornamental Shrubs and Hedges: Woody plants with multiple stems used for privacy, windbreaks, or visual structure.

Examples: Hibiscus, bougainvillea, ixora, privet.

Turfgrasses: Grasses cultivated to form a continuous ground cover for lawns, sports fields, and golf courses.

Examples: Bermuda grass, Zoysia grass, carpet grass, St. Augustine grass.

5. Spices, Condiments, and Medicinal Plants

These plants are cultivated for their aromatic, flavoring, or therapeutic properties. In tropical agriculture, they are a major sub-category of horticulture.

Spices and Condiments: Used to flavor or color food. Examples: Ginger, black pepper, turmeric, nutmeg, cloves.

Medicinal and Aromatic Plants: Cultivated for pharmaceutical or cosmetic extraction. Examples: Aloe vera, moringa, lemon grass, neem, mint.

6. Plantation Crops (Tropical Horticulture)

In tropical regions, many high-values, large-scale perennial crops are managed using intensive horticultural techniques, bridging the gap between horticulture and agronomy.

Beverage and Stimulant Crops: Examples: Cocoa, coffee, tea.

Industrial Horticulture: Cultivated for raw industrial materials. Examples: Rubber (latex), oil palm (though often classified broadly under agronomy, their nursery and plantation management heavily utilize horticultural science).

CLASSIFICATION OF HORTICULTURAL CROPS

Horticultural crops can be classified using several botanical and practical frameworks. Below is the classification based on their life-cycles, structural forms, and the specific parts of the plant utilized.

Classification Based on Life-Cycles

This classification categorizes plants based on the time it takes for them to complete their full biological life cycle, from seed germination to flowering, seed production, and eventual death.

Annuals: These plants complete their entire life cycle within a single growing season or one year. After producing fruit and seeds, the entire plant dies. Common examples include tomatoes, watermelons, spinach, sweet corn, and marigolds.

Biennials: These plants require two growing seasons to complete their life cycle. During the first year, they undergo vegetative growth (producing roots, stems, and leaves). In the second year, they utilize the stored energy to produce flowers, fruits, and seeds before dying. Common examples include carrots, onions, cabbage, and beets.

Perennials: These plants live for more than two years, and often for decades. Once they reach maturity, they typically flower and produce fruit every year without dying. Common examples include mangoes, apples, citrus trees, roses, and cashew trees.

Classification Based on Structures and Forms

This method categorizes plants based on their physical growth habits, stem woodiness, and overall morphology.

Herbs (Herbaceous Plants): These are plants with soft, fleshy, and non-woody stems. They usually die back to the ground at the end of the growing season. Common examples include lettuce, tomatoes, amaranth, and basil.

Shrubs: These are medium-sized, woody plants characterized by having multiple stems arising from or near the base of the plant. They lack a single main trunk and are generally shorter than trees. Common examples include hibiscus, bougainvillea, and coffee plants.

Trees: These are large, woody perennial plants featuring a single, distinct, and thick main stem (trunk) that supports a canopy of leaves and branches well above the ground. Common examples include sweet oranges, avocados, rubber trees, and oil palms.

Vines and Climbers: These plants possess weak, flexible stems that cannot support their own weight upright. They grow by trailing along the ground or climbing over structures, trees, or trellises using tendrils or twining stems. Common examples include grapes, passion fruit, cucumbers, and yams.

Classification Based on Uses and Parts Used

Because horticultural plants are highly diverse, they are often grouped by the specific anatomical part of the plant that is harvested for human consumption, industrial use, or aesthetic value.

Part of Plant Used Typical Category Common Examples
Roots Root Vegetables Carrots, radishes, sweet potatoes, beets
Tubers (Underground Stems) Tuber Crops Yams, Irish potatoes
Bulbs Bulb Vegetables Onions, garlic, leeks
Leaves Leafy Vegetables / Herbs Spinach, cabbage, lettuce, mint, tea
Stems Stem Vegetables Asparagus, sugarcane, kohlrabi
Flowers Floral Vegetables / Ornamentals Cauliflower, broccoli, roses, hibiscus
Fruits Fruit Vegetables / Tree Fruits Tomatoes, peppers, mangoes, apples
Seeds and Nuts Nut Crops / Legumes Cashews, walnuts, green peas, sweet corn
Latex / Sap Industrial Horticulture Rubber (Hevea brasiliensis)

Classification of Horticultural Crops

From time to time, horticultural crops have been classified into various groups depending on their growth habits, cultivation requirements, climate needs and uses. Horticultural crops are popularly classified into 3 broad groups of fruits and nuts, vegetables and flowers or ornamentals.

Classification Base on Use of Horticultural Plants

Vegetables Vegetables grown for the aerial edible parts (above the soil), Vegetables that have curled leaves e.g. Cabbage, cauliflower

Pulse crops – Vegetables that produce pods e.g. pea, bean or green bean.

Solanaceous fruits – Vegetables that produce fruits and tuber e.g. tomato, pepper, garden eggs, potato
Green of leafy Vegetables – Vegetables, that produce leafy e.g. Amaranth, bitter leaf, water leaf, spinach.
Salad crops – Vegetables that produce their product in vine e.g. melon, cucumber, pumpkin.

Corn vegetables e.g. popcorn, sweet corn. Vegetables or plants grown for the underground edible portion or parts.

Fruits Root crops e.g. carrot, potatoes, Irish, Cassava.

Tuber crops e.g. yam, cocoyam.

Bulbs crops e.g. onion, garlic.

Temperate fruits – fruits from cooler regions of the world
Tree fruits e.g. apple, peach.

Small fruits e.g. grapes vine, straw berry. **Nut fruits** e.g. peach, walnut.

Tropical and subtropical fruits (Hot regions) of the world
Tree fruits e.g. citrus, mango, guava, cashew
Herbaceous perennial fruits e.g. banana, plantain, pineapple.

Nut fruits e.g. cashewnut, datenut, arcanut. **Flowers and Ornamentals**
Flowers e.g. rose flower, marigold, sun flower
Lawns e.g. carpet grass, bahama grass
Hedges e.g. gambogi, croton yellow
Trees e.g. Christmas tree, umbrella tree, neem tree

Classification Based on Growth Habit

This is according to the growth nature of the crops and this include. Herbs e.g. Ageratum. Shrubs e.g. Hibiscus. Trees e.g. Mango, citrus. Climbers e.g. Bougainvillea. Creepers e.g. Bignonia, gracilis
Classification Base on Life Cycle of the Crop
Annual or seasonal crops – Those that complete their life cycle within a year e.g. marigold, Amaranth, tomato, Irish.
Biennial crops – Those that requires two years or at least two growing season to complete their life cycle e.g. Cassava, yam.
Perennial crops – Any plant that lives for more than two years to 35 years e.g. mango, citrus, grape vine, guava.

Classification Base on Climatic Requirement This is based on the temperature need of the crop and this are grouped into
Temperate crops – These are horticultural crops found in cold regions of the world e.g. apple, almond etc.
Tropical crops – These are horticultural crops that do not tolerate severe cold but can tolerate warm temperature e.g. banana, papaya, pineapple. Sub-

Tropical crops – They need warmth and humidity and can tolerate mild winter e.g. fig, mango, cashew nut

THE PROCESS OF SEED GERMINATION

Germination is the physiological process by which a dormant seed develops into a living, actively growing seedling under favorable conditions. It marks the transition from a resting state to an active vegetative growth phase.

Stages of Germination

1. **Imbibition (Water Absorption):** The seed absorbs water rapidly through its seed coat (testa) and the micropyle. This causes the seed to swell, softening the seed coat and eventually causing it to rupture.
2. **Enzyme Activation and Respiration:** The absorbed water activates dormant enzymes within the seed. Cellular respiration increases dramatically, utilizing oxygen to convert stored food reserves (such as the starchy endosperm found in maize) into soluble, usable energy.
3. **Embryo Growth:** The cells of the embryo begin to divide and elongate.
4. **Emergence:** The radicle (primary root) is typically the first structure to emerge, anchoring the plant and absorbing water from the soil. Shortly after, the plumule (primary shoot) emerges and grows upward towards the light to begin photosynthesis.

Types of Germination

Epigeal Germination: The cotyledons are pushed above the ground as the hypocotyl elongates. As they emerge, they often turn green and perform early photosynthesis. Example: Cashew.

Hypogeal Germination: The cotyledons remain below the soil surface while the epicotyl elongates to push the plumule upward. Example: Maize, rubber.

Factors Affecting Seed Germination

For a seed to successfully germinate, both external (environmental) and internal (physiological) conditions must be met.

External (Environmental) Factors

Water (Moisture): Essential for imbibition and enzyme activation. However, excess water can waterlog the soil, leading to rot.

Oxygen (Aeration): Germinating seeds require high amounts of oxygen for aerobic respiration to generate the energy needed for rapid cellular growth.

Temperature: Seeds have an optimum temperature range for the enzymatic reactions of germination. Extreme cold can halt the process, while extreme heat can denature the enzymes.

Light: While many agronomic crops germinate well in darkness, some specific weed and horticultural seeds require exposure to light to trigger germination (photodormancy).

Internal (Physiological) Factors

Seed Viability: The embryo must be alive and healthy. Dead or aged embryos cannot germinate.

Seed Dormancy: Some viable seeds will not germinate even when environmental conditions are perfect. This is a survival mechanism. Dormancy can be caused by:

Hard Seed Coats: Prevents water or oxygen entry.

Chemical Inhibitors: Presence of germination-inhibiting hormones.

Immature Embryos: Requires an "after-ripening" period.

Note: Certain plantation crops, like oil palm, have deep seed dormancy requiring specialized pre-treatments (like controlled heat chambers) to break the dormancy before sowing.

Qualities of Good Seeds

The success of any crop production endeavor, whether for an experimental research design or a commercial plantation, begins with high-quality planting material. A "good" seed lot must possess the following **characteristics**:

High Germination Percentage (Viability): A high proportion of the seeds must be alive and capable of germinating under normal conditions. Good quality seeds typically have a germination rate of 85% to 95% or higher.

High Seed Vigor: Vigor refers to the seed's ability to germinate rapidly and uniformly, and to develop into robust seedlings even under suboptimal or stressful field conditions.

Genetic Purity (True to Type): The seeds must genetically represent the specific variety or cultivar intended. There should be no mixing with other varieties, ensuring the crop exhibits the expected yield, disease resistance, and physical traits.

Physical Purity: The seed lot must be clean and free from inert matter (stones, broken seeds, chaff, soil) and, most importantly, free from the seeds of weeds or other crop species.

Health and Freedom from Pathogens: Good seeds are free from seed-borne diseases, fungal spores, and insect pest damage.

Optimum Moisture Content: To maintain viability during storage and prevent premature sprouting or fungal rot, seeds must be dried to a specific, safe moisture level before packaging.

Uniform Size and Weight: Plump, well-filled, and uniform seeds generally contain more food reserves, leading to stronger initial seedling establishment.

Seed Viability, Pre-Conditioning, and Factors Affecting Seed Health

I. Methods of Testing Seed Viability

Seed viability determines whether the embryo within a seed is alive, healthy, and capable of developing into a viable seedling. Testing ensures that poor seed lots are identified before planting.

1. Standard Germination Test (Direct Method)

Procedure: A known sample of seeds (e.g., 100 seeds) is placed on a moist substrate (blotting paper, paper towel, or sand) under controlled temperature and humidity conditions.

2. Tetrazolium (TZ) Test (Biochemical/Rapid Method)

Principle: Uses the chemical 2,3,5-triphenyl tetrazolium chloride. Living seed tissues undergo cellular respiration using dehydrogenase enzymes. These enzymes reduce the colorless TZ salt solution into an insoluble red compound called formazan.

Procedure: Seeds are soaked in water to soften, cut or bisected longitudinally to expose the embryo, and submerged in a 0.1–1% TZ solution in the dark. **Advantage:** Extremely fast (results available in 2–24 hours).

3. Embryo Excision Test

Procedure: Used primarily for slow-germinating tree or woody plant seeds with hard coats or deep dormancy. The embryo is carefully extracted from the seed coat and endosperm under sterile conditions and placed on moist filter paper.

Assessment: Viable embryos will green up, swell, or begin growing within a few days; dead embryos will turn brown, decay, or remain inactive.

4. Electrical Conductivity (EC) Test

Principle: Measures the integrity of cell membranes. As seeds age or deteriorate, cell membranes weaken, allowing electrolyte chemicals (potassium, sugars, amino acids) to leach out into water.

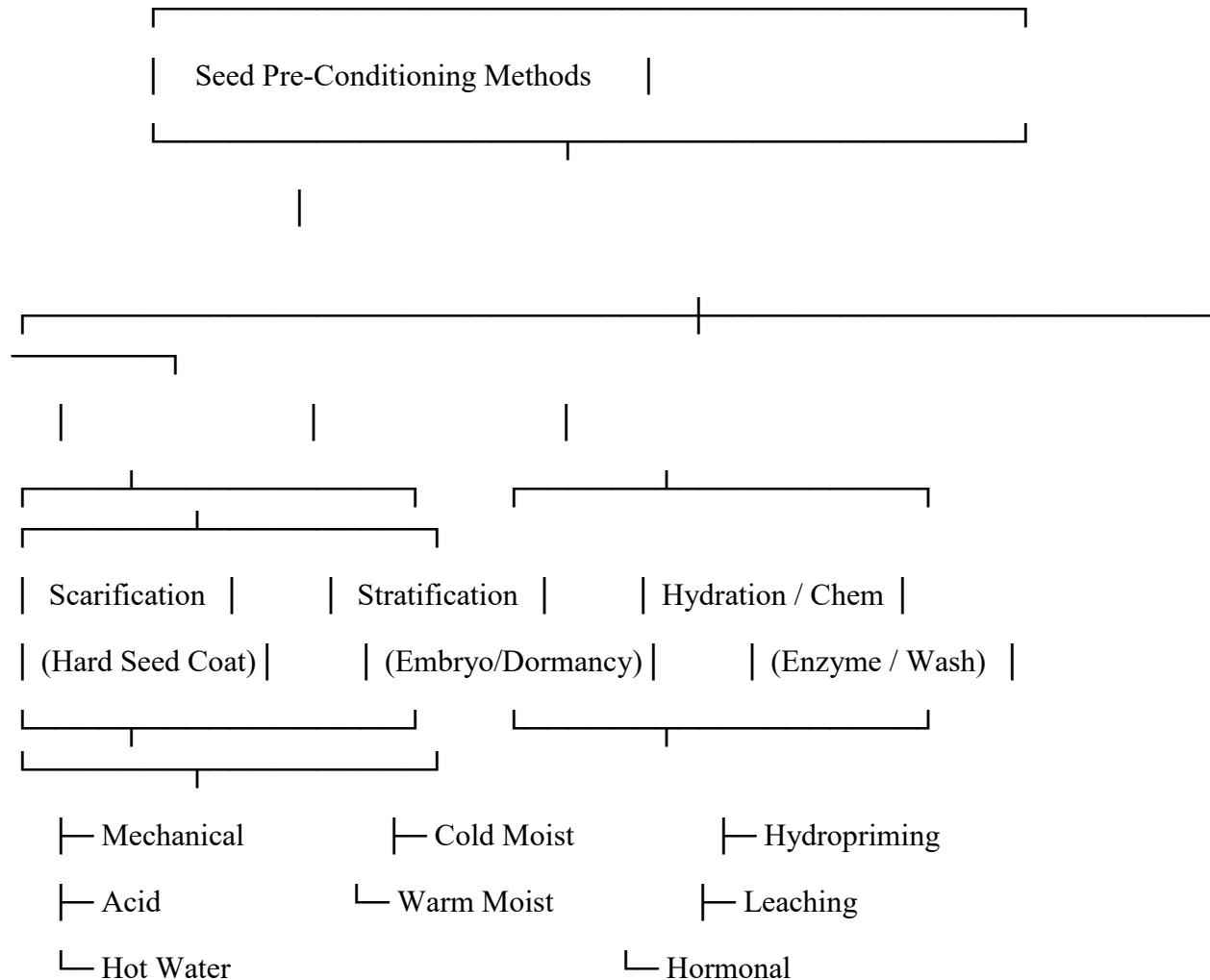
Procedure: Seeds are soaked in distilled water for a specific duration, and an EC meter measures the leachates in the water.

Assessment: High electrical conductivity indicates damaged membranes and low seed viability; low conductivity indicates intact cell walls and high seed viability.

Pre-Conditioning Seeds to Stimulate Germination

Pre-conditioning involves treatments applied prior to sowing to break seed dormancy, speed up germination, and promote uniform seedling emergence.

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1. **Scarification** (Breaking Physical Dormancy)

Used when a hard, impermeable seed coat prevents water or oxygen entry.

Mechanical Scarification: Physical wearing, nicking, or scratching of the seed coat using sandpaper, files, or mechanical tumblers.

Acid Scarification: Soaking seeds in concentrated sulfuric acid (H_2SO_4) for a specific duration (1 minute to over an hour) to etch the outer coat, followed by thorough washing.

Hot Water Treatment: Immersion of seeds in hot water (70°C–100°C) for a short period, then letting them soak in cooling water for 12–24 hours.

2. Stratification (Breaking Physiological Dormancy)

Exposing seeds to specific temperature and moisture conditions to mimic seasonal changes required for embryo maturation.

Cold Moist Stratification: Storing imbibed (moist) seeds in a refrigerated environment (2°C–5°C) for several weeks to months. (Commonly needed for temperate fruit trees).

Warm Stratification: Exposing seeds to warm, moist conditions (20°C–30°C) prior to cold treatment to soften tissues or allow embryo development.

3. Priming and Hydration Treatments

Hydropriming (Soaking): Soaking seeds in water for 12–24 hours prior to sowing to initiate hydration and early metabolic activity without permitting the radicle to break through.

Osmopriming: Soaking seeds in osmotic solutions (such as Polyethylene Glycol - PEG) to control water uptake rate precisely.

4. Chemical and Hormonal Treatments

Chemical Leaching: Washing seeds in running water for 12–48 hours to remove germination-inhibiting compounds (like abscisic acid) from the seed coat.

Growth Regulators: Treating seeds with phytohormones like Gibberellic Acid

Factors Affecting the Viability of Seeds

Seed viability degrades over time, but the rate of deterioration is governed by both intrinsic traits and storage environments.

A. Internal (Genetic & Pre-Harvest) Factors

Genetic Makeup / Species Specificity: Different species inherently possess different natural lifespans. (e.g., short-lived recalcitrant seeds like rubber or cocoa vs. long-lived orthodox seeds like legumes).

Initial Seed Quality & Maturity: Seeds harvested before full physiological maturity possess underdeveloped embryos and lower food reserves, resulting in rapid loss of viability.

Mechanical Damage during Harvest: Cracks in the seed coat or mechanical stress sustained during threshing create points of entry for pathogens and accelerate degradation.

B. External (Storage Environment) Factors

Seed Moisture Content (SMC):

High moisture levels (>14%) increase respiration rates, cause heating, and encourage fungal and pest development.

Extremely low moisture levels (<4%) can cause membrane structural damage in some species.

Storage Temperature: High temperatures accelerate internal metabolic degradation, chemical oxidation, and lipid rancidity. Cold storage slows respiration and prolongs viability.

Relative Humidity (RH) of Storage: Seeds are hygroscopic (they absorb moisture from the surrounding air until equilibrium is reached). High storage RH directly elevates seed moisture content.

Oxygen Levels: High oxygen levels promote oxidative stress and aerobic respiration, shortening storage life. Gas-tight or hermetic containers with low oxygen help preserve vigor.

Pathogens and Storage Pests: Fungal spores (e.g., *Aspergillus*, *Penicillium*) and insects (e.g., weevils) digest endosperm reserves, generate heat, and rapidly destroy embryo viability.

> Harrington's Rules of Thumb for Seed Storage:

> 1. For every 1% reduction in seed moisture content (between 5% and 14%), the storage life of the seed is doubled.

> 2. For every 5°C reduction in storage temperature (between 0°C and 50°C), the storage life of the seed is doubled.

Sowing Methods: In Situ vs. Nursery Sowing

Once seeds have been selected, tested for viability, and pre-conditioned if necessary, the next critical step in crop production is sowing. The method chosen depends on the plant species, seed size, environmental conditions, and the scale of the agricultural operation.

There are two primary methods of sowing seeds: In situ (direct sowing) and Nursery sowing (indirect sowing).

A. Sowing In Situ (Direct Sowing)

Definition: Sowing in situ (a Latin phrase meaning "on site" or "in position") refers to planting seeds directly into the permanent field or garden where the plant will grow, mature, and be harvested.

This method is typically used for crops with large seeds that contain sufficient food reserves to survive early field conditions, or for crops that do not tolerate root disturbance during transplanting.

Common Methods of Direct Sowing:

1. **Broadcasting:** Seeds are scattered evenly over the surface of the prepared soil by hand or machine, followed by light raking to cover them. (Common for turfgrasses, cover crops, and some small-seeded leafy vegetables).
2. **Dibbling:** Making small holes (using a dibber or stick) at specific distances and placing one or more seeds into each hole before covering. (Common for maize, beans, and melon).
3. **Drilling (Row Planting):** Dropping seeds in continuous, shallow furrows (drills) at specific row spacing and covering them with soil. (Common for commercial field crops like wheat or soybeans).

Advantages of In Situ Sowing:

Cost and Labor Efficient: Eliminates the labor, time, and materials required to manage a nursery and later transplant seedlings.

No Transplant Shock: Plants establish their taproots directly in the final soil, avoiding the growth stunt or root damage often associated with transplanting.

Faster Field Operations: Large areas can be mechanized and sown rapidly using tractor-drawn planters.

Disadvantages of In Situ Sowing:

High Seed Wastage: Requires a larger volume of seeds to compensate for poor germination, pests, and erratic weather in the open field.

Uneven Stand Establishment: Field conditions (heavy rain, bird attacks, soil crusting) are harder to control, which can lead to patchy crop emergence.

Weed Competition: Young, delicate seedlings must immediately compete with aggressive field weeds for light, water, and nutrients.

B. Sowing in the Nursery (Indirect Sowing)

Definition: Nursery sowing involves planting seeds in a specially prepared, protected, and intensively managed area (the nursery) to raise seedlings. Once the seedlings reach a specific stage of maturity and hardiness, they are uprooted and transplanted to their permanent field location.

This method is essential for crops with very small, delicate seeds, expensive hybrid seeds, or slow-growing perennial crops.

Types of Nursery Receptacles:

1. **Raised Seedbeds:** Mounded beds of fine, nutrient-rich soil that provide good drainage and ease of management.
2. **Seed Trays / Plug Trays:** Plastic or polystyrene trays with individual cells, allowing for easy extraction of the root ball with minimal disturbance.
3. **Polythene Bags (Poly-bags):** Frequently used for tree and plantation crops. Seeds are sown into individual soil-filled bags, allowing them to grow for several months before field planting.

Advantages of Nursery Sowing:

Optimal Environment: Seedlings are protected from harsh weather (heavy rain, intense sun) and field pests. Shade nets and controlled irrigation ensure high germination rates.

Economy of Seeds: Because conditions are optimal, fewer seeds are wasted. This is highly cost-effective for expensive hybrid or genetically modified seeds.

Selection of Vigorous Seedlings: Only the healthiest, most robust seedlings are selected for transplanting, ensuring a uniform and highly productive field stand.

Better Land Utilization: While the seedlings are growing in the nursery (which takes up a fraction of the space), the main field can be prepared, or a preceding crop can finish its harvest cycle.

Disadvantages of Nursery Sowing:

Transplant Shock: The process of moving seedlings can damage fragile root hairs, causing temporary wilting and a pause in growth until the plant recovers. (Poly-bags and plug trays mitigate this).

Highly Labor Intensive: Requires intensive daily management (watering, weeding, shading) and skilled labor for the transplanting process.

Higher Initial Cost: Requires investment in nursery infrastructure, such as shade structures, irrigation equipment, poly-bags, and specialized potting media.

Feature In Situ (Direct Sowing) Nursery (Indirect Sowing)
Typical Seed Size Large, robust seeds Very small or highly expensive seeds
Common Crop Examples Maize, Beans, Carrots, Okra, Radish Tomatoes, Peppers, Cocoa, Oil Palm,
Resource Intensity Lower initial labor/capital High initial labor/capital
Environmental Control Low (Subject to field weather) High (Shade, precise watering)
Transplant Shock Risk None Moderate to High (Requires careful handling)
Land Preparation Timing Must be ready before sowing Can be prepared while seedlings grow

THE CONCEPT AND OBJECTIVES OF SOIL STERILIZATION

Soil sterilization (more accurately referred to in agriculture as soil pasteurization) is the process of subjecting potting soil or seedbed media to physical or chemical treatments prior to planting. The goal is to eradicate or severely suppress populations of harmful biological agents that reside in the soil, creating a clean environment for delicate seeds and young plants.

Objectives of Soil Sterilization:

1. **Eradication of Soil-Borne Pathogens:** To destroy harmful fungi, bacteria, and viruses that cause diseases such as "damping-off" (a fungal disease that rots seedlings at the soil line), root rots, and vascular wilts.
2. **Destruction of Weed Seeds:** To kill dormant weed seeds in the soil, ensuring that the crop seedlings do not face competition for light, water, and nutrients in their vulnerable early stages.
3. **Control of Nematodes and Soil Insects:** To eliminate root-knot nematodes, wireworms, grubs, and other microscopic or macroscopic soil pests that damage plant roots.
4. **Improved Seedling Vigor:** By removing biological competition and pathogens, crop seeds experience higher germination rates, and seedlings grow more vigorously and uniformly.

Methods of Sterilizing Soil

While there are several methods (including solarization and dry heat), the two primary industrial and horticultural methods are steam and chemical sterilization.

1. **Steam Sterilization (Physical Method)**

This is widely considered the safest and most effective method for greenhouse and nursery operations. It involves passing hot steam through the soil mass.

Mechanism: Steam is injected into the soil until the temperature of the entire soil mass reaches approximately 60°C to 82°C for about 30 minutes. This temperature range is sufficient to kill most pathogenic fungi, bacteria, nematodes, and weed seeds without completely destroying beneficial, heat-resistant soil bacteria.

Advantages:

Leaves no toxic chemical residues.

Soil can be used for planting as soon as it cools down.

Highly effective and broad-spectrum.

Disadvantages:

Requires specialized, often expensive, boiler equipment.

Over-steaming (exceeding 90°C) can kill all beneficial microbes, leading to a biological vacuum where new pathogens can spread rapidly if re-introduced. It can also cause manganese toxicity in certain soils.

2. Chemical Sterilization (Fumigation / Drenching)

This method involves applying volatile toxic chemicals (fumigants) or liquid drenches to the soil to kill pests and pathogens.

Mechanism: Chemicals are applied to the soil, which is often then covered with a plastic tarp to trap the lethal gases. Common chemicals include:

Formaldehyde: Used as a liquid drench, primarily effective against fungi.

Dazomet (Basamid): A granular material that, when mixed with moist soil, releases toxic methyl

Methyl Bromide: Historically the most popular and powerful fumigant, but now largely banned globally due to its severe ozone-depleting properties.

Advantages:

More practical for large, open field applications where steaming is impossible.

Requires less specialized machinery compared to boilers.

Disadvantages:

Highly toxic to humans and the environment; requires strict safety gear and regulations.

Leaves residues. The soil must undergo a strict "aeration" or waiting period (often 2 to 3 weeks) before planting, otherwise, the chemical will kill the crop seeds (phytotoxicity).

Uses of Sterile Soil

Sterile (or pasteurized) soil is not typically used for large-scale field agronomy due to the sheer volume of land. Its use is highly targeted within horticulture and experimental agriculture:

1. **Nursery Seedbeds and Trays:** To raise delicate seedlings (like tomatoes, peppers, cocoa, and ornamental flowers) ensuring high survival rates before they are transplanted to the unsterile field.
2. **Greenhouse and Potted Plants:** High-value indoor plants require weed-free and pathogen-free media, as the warm, humid greenhouse environment is ideal for disease proliferation.
3. **Tissue Culture Acclimatization:** When plantlets are moved from completely sterile laboratory test tubes into the real world, they have zero natural immunity. Sterile potting mix is crucial for their survival during the weaning phase.
4. **Pathology and Agronomy Experiments:** Researchers must use sterile soil as a "control" medium to test specific plant-pathogen interactions or precise fertilizer responses without interference from unknown soil microbes.

Techniques for Handling Sterile Soil

The most critical aspect of using sterile soil is preventing re-contamination. Because the soil has been stripped of competing microbes, any introduced pathogen will multiply exponentially.

Sanitize Tools and Equipment: All pots, poly-bags, shovels, and watering cans must be washed and disinfected (often using a bleach solution) before coming into contact with the sterile soil.

Clean Water Source: Irrigation water must be clean. Using contaminated pond or river water will immediately re-infect the soil.

Elevated Benches: Keep seed trays and sterile soil bags off the bare ground (using wire or slatted benches) to prevent soil-borne pathogens and nematodes from splashing or crawling up into the sterile media.

Proper Storage: If not used immediately, sterile soil should be kept in sealed, clean bags or covered bins to prevent wind-blown weed seeds and fungal spores from settling on it.

ECONOMIC FRUITS AND FRUIT TREES IN NIGERIA

Fruit production is a cornerstone of Nigeria's agricultural economy, providing raw materials for the manufacturing sector, generating foreign exchange through exports, and supplying vital dietary nutrients.

Common Name Botanical Name Plant Family Primary Economic Product
Citrus (Sweet Orange) <i>Citrus sinensis</i> Rutaceae Fresh fruit, juice concentrate, essential oils
Mango <i>Mangifera indica</i> Anacardiaceae Fresh fruit, juice purée, dried fruit snacks
African Cherry (Agbalumo/Udara) <i>ChryIsophyllum albidum</i> Sapotaceae Fresh fruit, local jams, traditional medicine
Avocado Pear <i>Perslea americana</i> Lauraceae Fresh fruit, cosmetic oils, culinary oils
Cashew <i>Anacardium occidentale</i> Anacardiaceae Raw cashew nuts (export), roasted nuts, apple juice

Importance to the Nigerian Economy

Export and Foreign Exchange: Cashew is one of Nigeria's top non-oil exports, generating hundreds of millions of dollars annually. Mangoes and avocados also hold massive untapped export potential for European and Middle Eastern markets.

Agro-Industrial Development: Citrus and mango plantations directly feed indigenous fruit juice concentration plants, reducing the national import bill for fruit concentrates. Avocado is increasingly cultivated for oil extraction used in the booming local and international cosmetic industries.

Rural Wealth Creation: The African Cherry (Agbalumo/Udara) is a highly lucrative seasonal crop. Because it is largely harvested from the wild or semi-domesticated trees, it provides a massive, rapid influx of cash to rural gatherers and female market traders during its short season.

Nutritional Security: These fruits provide dense sources of Vitamin C (Citrus), Vitamin A (Mango), healthy monounsaturated fats (Avocado), and protein (Cashew nuts), fighting malnutrition across the country.

Varieties and Growth Requirements

To achieve commercial success, plantation managers must select the right variety and provide the specific environmental conditions required by each species.

1. **Citrus** (Focus: Sweet Orange - *Citrus sinensis*)

Common Varieties in Nigeria:

Agege Sweet: Adapted to the humid forest zones, highly popular in local markets.

Valencia: Late-season, high juice content, demanded by beverage processing factories.

Washington Navel: Seedless, easy-peeling, premium table fruit.

Growth Requirements:

Climate: Subtropical to tropical. Optimal temperatures are 21°C to 32°C. High humidity increases the risk of fungal diseases like Phytophthora foot rot.

Rainfall: Requires 1,000mm to 2,000mm annually, well-distributed.

Soil: Deep, well-drained sandy-loam or loamy soils (pH 5.5–6.5). Highly sensitive to waterlogging.

2. **Mango** (*Mangifera indica*)

Common Varieties in Nigeria:

Local (Ogbomoso/Sheri): Highly fibrous, very sweet, robust, and disease-resistant.

Alphonso & Kent: Low fiber, thick flesh, long shelf-life (ideal for export and processing).

Julie / Peter: Popular table varieties with excellent flavor and moderate fiber.

Growth Requirements:

Climate: Requires a tropical climate with a strict, distinct dry season (3–4 months) to induce heavy flowering and prevent rain from washing away pollen or causing anthracnose fungal rot on flowers.

Rainfall: 700mm to 1,500mm annually.

Soil: Deep, well-drained soils. Develops a massive taproot and can scavenge nutrients from poorer soils, but yields best on deep loams.

3. **African Cherry** (*Chrysophyllum albidum*)

(Note: True temperate cherries do not grow in Nigeria due to a lack of winter chill hours. In Nigerian agronomy, "Cherry" refers to the African Star Apple).

Common Varieties in Nigeria:

Currently, there are few formally bred varieties. The crop relies on local "landraces" generally categorized by consumers into "Sweet" and "Sour" types.

Growth Requirements:

Climate: Strictly adapted to the lowland tropical rainforest zones. Requires high humidity and warm temperatures year-round (25°C to 30°C).

Rainfall: Demands high rainfall, typically 1,500mm to 3,000mm annually.

Soil: Prefers acidic to neutral (pH 5.0–6.5), well-drained, organically rich forest soils. It is shade-tolerant when young but requires full sun for maximum fruiting.

4. **Avocado (*Persea americana*)**

Common Varieties in Nigeria:

Local Landraces: Often green-skinned, large, and watery (West Indian race characteristics).

Hass: Small, dark-purple bumpy skin, extremely high oil content, and the global standard for export.

Fuerte: Green-skinned, pear-shaped, cold-hardy, excellent flavor.

Growth Requirements:

Climate: Humid tropical to subtropical. Optimal temperatures range from 20°C to 28°C.

Rainfall: 1,200mm to 2,500mm annually. Needs consistent moisture but zero standing water.

Soil: Drainage is the single most critical factor. Avocados are highly susceptible to *Phytophthora cinnamomi* root rot in wet, heavy clay soils. They require deep, aerated, sandy-loam soils.

5. **Cashew (*Anacardium occidentale*)**

Common Varieties in Nigeria:

Local Varieties: Smaller nuts, low apple-to-nut ratio, highly resilient to drought.

Brazilian Jumbo: Improved variety with massive nuts, high market premium, and precocious (early-fruited) growth.

Growth Requirements:

Climate: A rugged tropical plant that thrives in heat (25°C to 35°C). Like the mango, it absolutely requires a dry season for the nuts to mature properly and to facilitate harvest (as nuts fall to the ground to be gathered).

Rainfall: 800mm to 1,200mm annually. Highly drought-resistant once the taproot is established.

Soil: Excellent for marginal lands. Will grow in poor, degraded, sandy, or lateritic soils where other crops fail, provided the soil is not prone to flooding or hardpan compaction.

Fruit is the sweet or fleshy seed-bearing product that grows from a flower, while a fruit tree is the perennial plant with a woody trunk and branches that produces that fruit. A fruit is the seed-containing part that grows from the ovary of a flowering plant. Can grow on trees, bushes, vines, or low-lying plants (e.g., strawberries, watermelons), Protects seeds and helps scatter them so new plants can grow. A fruit tree is a large, woody plant that lives for many years and bears edible or useful fruits, Consists of roots, a main trunk, branches, leaves, and flowers. Examples: Apple trees, mango trees, and orange trees.

African Star Apple: Known locally as agbalumo (Yoruba) or udara (Igbo).

Black Velvet Tamarind: Popularly called awin (Yoruba) or icheku (Igbo).

Hog Plum: Known as iyeye (Yoruba) or ngulungwu (Igbo).

Common Cultivated Fruits

Pawpaw (Papaya): Available year-round.

Watermelon: Peaks during the dry season.

Oranges and Tangerines: Part of the local citrus family (osan).

Guava: Rich in vitamin C (goba).

Soursop: Valued for its creamy, sweet pulp.

Other fruit that are not fruit tree

Several common fruits grown or eaten in Nigeria grow on large herbs, vines, or low plants rather than actual woody trees, including watermelon, pineapples, and bananas.

Vine and Ground Fruits-

Watermelon (Bara / Elegede): Grows on long creeping vines that spread directly across the ground.

Pineapple (Ope oyinbo): Grows on a short, spiky, herbaceous plant near the soil level rather than on a trunk.

Cucumber: Grows on trailing or climbing vines, acting as a fruit botanically while coming from a non-woody plant.

Herbaceous and Shrub Fruits

Banana and Plantain (Ogede / Ayaba / Unere): Grow on large, pseudo-stemmed herbaceous plants that look like trees but are actually the world's largest herbs.

Strawberry: Grows on low-lying green runner plants close to the ground, cultivated mostly in cooler highland areas like Jos.

Tomatoes: Botanically classified as fruit berries and produced widely across Nigerian farms on small branching bush plants.

PRINCIPLES AND PRACTICES OF ORCHARD ESTABLISHMENT

Establishing a commercial orchard for perennial fruit trees—such as Citrus, Mango, African Cherry, Avocado, and Cashew—is a long-term capital investment. Errors made during the establishment phase will permanently affect the orchard's productivity and lifespan. Therefore, strict adherence to agronomic principles is mandatory.

i. Raising of Seedlings in the Nursery

Perennial fruit trees are rarely sown directly in the field due to their delicate early growth stages and the need for genetic uniformity.

(Note: While the curriculum mentions "suckers," none of the specified crops—Citrus, Mango, Cherry, Avocado, or Cashew—are commercially propagated via suckers. Suckers are used for crops like Plantain, Banana, and Pineapple. The tree crops listed rely on seeds and vegetative grafting).

Nursery Practices:

Containers: Seeds are typically sown into large, perforated polythene bags (poly-bags) filled with sterilized, nutrient-rich topsoil. This allows for safe transport and minimizes root damage during transplanting.

Propagation Methods by Crop:

Citrus: Seeds of hardy rootstocks (e.g., Rough Lemon or Cleopatra Mandarin) are sown. Once established, they are budded (T-budding) with the desired sweet orange variety (scion) to ensure disease resistance and fruit quality.

Mango & Avocado: Seeds are sown to grow a rootstock. They are later cleft-grafted or whip-grafted with superior commercial varieties (e.g., Alphonso mango or Hass avocado). Grafting ensures the trees are true-to-type and fruit much earlier than seed-grown trees.

Cashew: Because cashew has a highly sensitive taproot, selected large, heavy nuts (which indicate high vigor) are planted directly into deep poly-bags. Grafting is becoming more common for improved varieties (like Brazilian Jumbo).

African Cherry (*Chrysophyllum albidum*): Typically propagated by seeds extracted from healthy, sweet-tasting parent trees and raised in poly-bags under shade nets.

Hardening Off: About a month before transplanting, shade is gradually reduced and watering is decreased to acclimatize the seedlings to harsh field conditions.

ii. Land Clearing

The objective of land clearing is to prepare a clean, manageable environment that eliminates competition from native vegetation and removes sources of pests and diseases.

Under-brushing and Felling: Removing shrubs, vines, and felling native timber.

Stumping (Crucial Step): Unlike annual crops, orchards require the complete removal of native tree stumps and large roots. If left to rot in the soil, old stumps become breeding grounds for root-rot fungi (such as *Armillaria mellea* or *Fomes* spp.) and termites, which will eventually spread to and kill the newly planted fruit trees.

Plowing and Harrowing: Optional, but highly recommended if the farmer plans to grow short-term catch crops (like maize or cowpea) between the tree rows during the early years of the orchard.

iii. Marking-Out

Marking-out is the process of transferring the orchard design from a paper plan onto the cleared land. It ensures trees are planted in straight lines at precise distances.

Tools Used: Ropes, measuring tapes, ranging poles, and wooden pegs.

Planting Systems:

Square System: The most common. Trees are planted at the corners of a square, making tractor movement easy in both directions.

Triangular / Quincunx System: Allows for more trees per hectare and better utilizes sunlight.

Standard Spacing (Approximate):

Citrus: 7m x 7m

African Cherry: 8m x 8m

Avocado: 8m x 8m to 10m x 10m (Depending on variety and pruning strategy)

Mango & Cashew: 10m x 10m (These develop massive canopies and require extensive room).

iv. Holing

Holing involves digging the actual planting pits where the pegs were placed during marking-out. This should be done several weeks before the onset of the heavy rains.

Dimensions: A standard planting hole for these tree crops is 60cm x 60cm x 60cm (length x width x depth).

Soil Separation: When digging, the fertile topsoil (the first 15–20cm) must be separated from the less fertile subsoil.

Refilling: The topsoil is mixed with well-rotted organic matter (compost or poultry manure) and a basal dose of phosphorus fertilizer (like Single Super Phosphate, which aids early root development). This enriched topsoil is then placed back into the bottom of the hole. The subsoil is left on the surface to form a water-catchment ring later. Refilling early allows the soil to settle before the seedling arrives.

v. Transplanting

Transplanting is the physical transfer of the seedling from the nursery to the field. It is a delicate operation usually performed early in the rainy season to ensure natural moisture availability.

Transplanting Steps:

1. **Watering:** Water the seedlings in the nursery the evening before transplanting so the soil holds together in a cohesive root ball.

2. **Bag Removal:** Carefully cut away the poly-bag with a sharp knife. Never pull the plant out by the stem, and ensure the root ball does not shatter. An intact root ball prevents transplant shock.

3. **Planting Depth:** Place the root ball into the center of the refilled hole. The seedling must be planted at the exact same depth it was growing in the nursery.

> **Crucial Rule for Grafted Trees:** For Mango, Citrus, and Avocado, the graft union (the swollen joint where the scion meets the rootstock) must remain at least 15cm above the soil line. If buried, the scion will develop its own roots, or soil-borne fungi will rot the joint.

4. **Firming:** Gently press the soil around the root ball to eliminate air pockets, which can dry out the roots.

5. **Watering and Mulching:** Water the plant immediately. Apply a thick layer of dry grass (mulch) around the base (keeping it a few inches away from the trunk to prevent collar rot) to conserve moisture and suppress weeds.

6. **Staking:** If the seedlings are tall and exposed to wind, tie them loosely to a bamboo stake using soft twine.

Maintenance Practices in Established Orchards

Once an orchard is successfully established, it transitions from the vegetative growth phase into its reproductive (fruiting) lifespan, which can last for several decades. To sustain high economic

yields and prolong the life of fruit trees like Citrus, Mango, Cashew, Avocado, and African Cherry, rigorous routine maintenance is mandatory.

i. Weeding (Orchard Floor Management)

Weeds compete fiercely with fruit trees for soil moisture, essential nutrients, and sunlight (especially in young trees). Furthermore, dense weed undergrowth creates a humid microclimate that favors fungal diseases and provides a breeding ground for pests (like rodents and grasshoppers).

Ring Weeding: The most critical practice is keeping the base of the tree entirely weed-free. A clean circle (ring) of about 1 to 2 meters in radius around the trunk should be maintained manually or with careful herbicide use. This prevents collar rot and protects the trunk from rodent attacks.

Inter-row Slashing: Complete bare soil in the entire orchard can lead to severe soil erosion. Instead, the vegetation between the tree rows should be periodically slashed (using tractor-mounted slashers or cutlasses) to a low height.

Cover Cropping: To suppress noxious weeds and improve soil health, leguminous cover crops (such as *Centrosema* or *Mucuna*) are often planted between rows.

Chemical Control: Herbicides (like Glyphosate) can be used, but extreme care must be taken to ensure the chemical spray does not drift onto the green bark or low-hanging leaves of the fruit trees, as this can cause severe phytotoxicity.

ii. Pruning

Pruning is the selective removal of specific plant parts (branches, buds, or roots) to manipulate the tree's architecture, improve airflow, and enhance fruit production.

Sanitary Pruning (All Species): The universal removal of dead, diseased, broken, or parasitic branches (such as mistletoe infestations common in Citrus and African Cherry).

Canopy Management by Crop:

Mango: Requires "center-opening" pruning. The inner branches are removed to create a vase-like structure, allowing sunlight to penetrate the canopy and air to circulate. This significantly reduces the incidence of Anthracnose fungal disease.

Citrus: Involves "skirt pruning"—removing low-hanging branches that touch the ground to prevent *Phytophthora* pathogens from splashing from the soil onto the fruits. "Water sprouts" (vigorous, non-fruiting vertical shoots) are also removed.

Cashew: Generally requires less intensive structural pruning than mango or citrus. However, lower branches must be pruned upward. Because mature cashew nuts fall to the ground for harvesting, the area beneath the canopy must be accessible and clear.

Avocado: Primarily pruned to control height. Tall, unpruned avocado trees make manual harvesting incredibly difficult and dangerous.

iii. Diseases and Pest Control

An Integrated Pest Management (IPM) approach—combining biological, cultural, and chemical controls—is necessary to protect the economic value of the fruits.

Common Orchard Pests:

Fruit Flies (*Bactrocera* spp. & *Ceratitis* spp.): Devastating to Mango and Citrus. Females lay eggs inside the maturing fruit; the larvae hatch and cause the fruit to rot and drop prematurely. Control: Use of pheromone traps, bait sprays, and strict orchard sanitation (burying all fallen, rotting fruits).

Cashew Stem Girdlers and Thrips: Thrips severely damage young cashew inflorescences, reducing nut set. Control: Targeted insecticide applications during the flushing and flowering stages.

Common Orchard Diseases:

Anthrachnose (*Colletotrichum gloeosporioides*): A severe fungal disease affecting Mango, Cashew, and Avocado. It causes black, sunken lesions on leaves, flowers, and fruits, often destroying the entire blossom (blossom blight). Control: Pruning for airflow and application of copper-based fungicides before and during flowering.

Phytophthora Root and Collar Rot: Particularly lethal to Avocado and Citrus in poorly drained soils. Control: Ensuring excellent soil drainage, avoiding trunk injury during weeding, and using resistant rootstocks.

iv. Fertilizer Application and Manuring

Fruit trees export massive amounts of nutrients from the soil via the harvested fruits. These nutrients must be systematically replaced to prevent yield decline, alternate bearing (producing a heavy crop one year and nothing the next), and premature tree death.

Organic Manuring and Nitrogen Fertilization: Integrating organic nitrogen fertilization is crucial for long-term orchard health, particularly for robust canopy development in crops like cashew and mango. Applying well-composted poultry manure or farmyard manure not only provides a slow, steady release of organic nitrogen but also rebuilds soil organic matter, improves water retention, and stimulates beneficial soil microbes.

Inorganic Fertilizers: Standard NPK (Nitrogen, Phosphorus, Potassium) blends are applied to provide immediate, targeted nutrient boosts.

Nitrogen is vital for vegetative flushes.

Phosphorus supports root health and flower initiation.

Potassium is critical during fruit development for size, sugar content, and disease resistance.

Placement Method (Ring Application): The active, nutrient-absorbing feeder roots of a mature fruit tree are located at the outer edge of the leaf canopy (the drip line). Fertilizers and manure should never be piled against the trunk. Instead, they should be applied in a shallow trench dug in a ring around the tree at the drip line, and then lightly covered with soil to prevent volatilization and runoff.

Timing: Nutrient applications are typically split into two doses: the first at the onset of the rainy season to support the new vegetative flush and flowering, and the second towards the end of the rainy season to support fruit maturation and post-harvest recovery.

PLANT PROPAGATION

Plant propagation is defined as the controlled process of creating new plants from a variety of sources such as seeds and other plant parts.

The objectives of plant propagation are to:

- i. Increase in number of plants
- ii. Preserve the essential characteristics of plant
Propagation of crop involves the formation and development of new individuals using specialized part of the plant. These new individuals are used in the establishment of new plantings.

Methods of Propagation of Horticultural Crops

In general, two methods are employed in propagation of horticultural crop

1. The use of seed (sexual)
2. The use of vegetative part of plants (asexual).

Sexual Propagation (seed) This is a method of increasing the number of plants through seeds formed from the union of gametes during pollination. Essentially, a seed consists of an embryo with nourishing and protecting tissue. The embryo is considered a minute plant. Principal parts of the embryo are plumule (which is the first growing point of the stem), the radical (the first growing point of the root) and the hypocotyls and epicotyls (together constitute the first, or original, stem of the plant). The nourishing tissues are endosperm and cotyledons. In well-

developed mature seed, these tissues are packed with stored food such as starch, hemicelluloses, reserved proteins, or fats depending on the kind of plant. The protecting tissue is the seed coat. In general, the seed coat retards the rate of transpiration; in some kinds of plants, it retards the rate of respiration while the seeds are in storage and protects the delicate embryo from mechanical injury to some extent. When the seed is exposed to environmental condition favourable for its growth (soil, water, temperature, light etc), the seed germinates and produces a plant. Therefore, a seed may be defined as a minute plant with nourishing and protecting tissues that can be used to produce a similar plant. Sexual mode of propagation is the principal means by which biological variation is generated.

Advantages of sexual propagation

Easy and natural phenomenon – once the seed is given the right environmental condition it will germinate and produce the plants.

They are sources of genetic variability – They provide means of recombination of genetic material to produce improved varieties.

It helps in selecting different traits – The desirable traits can be retained while undesirable traits removed.

Disadvantages of sexual propagation

Not all plants produce viable seeds e.g. banana pineapple.

It is not possible to maintain a true – to – type genotype

Sexual method of propagation takes longer period of time from seed to harvest. This is especially true with respect to tree crops.

Asexual Propagation (Vegetative Propagation)

Vegetative propagation is the method that uses any part of plant rather 'true' seed to produce new plants. Plants have a number of vegetative mechanisms. Some of these have been taken advantage of by horticultural and gardeners to multiply or clone plants rapidly. Plants produced using vegetative parts have no exchange of genetic material; therefore, the plants are identical to the parent. Vegetative propagation uses plant parts such as roots, stem and leaves. In some plants, seeds can be produced without fertilization and the seeds contain only the genetic material of the parent plant. This is called apomixes. Apomixes is asexual reproduction but not vegetative propagation. Vegetative propagation is essential for the raising of many economic crops eg fruit crops, nut crops, many flowering and ornamental crops and certain Vegetables crops.

Advantages of Vegetative Propagation

Maintenance of true – to – type clones over a time.

A clone is a group of plants with the same genetic makeup.

It maintains uniformity of plants genotype.

It is the only means of propagating certain plants e.g. banana and plantain which do not produce viable seeds.

Earliness of production is achieved.

It may be more economical to propagate asexually in some crops as in sweet potatoes.

Some seeds germinate with difficulty and as such vegetative propagation is best for them.

Seed borne diseases are avoided when plants are propagated asexually.

Disadvantages of Vegetative Propagation

Planting materials are usually bulky.

Storage of asexual material is cumbersome and usually short term.

It does not provide source of genetic variability for recombination of gene.

Mechanized propagation in most cases is not practicable.

Need specialised personnel to handle the procedures.

It is expensive than seed.

Systemic viral diseases can spread to all plants through plant materials and tools used.

The methods of vegetative propagation include the following;

- a. Use of specialised vegetative structure eg. Root, corms, bulb, rhizomes etc.
- b. Cutting
- c. Layering
- d. Grafting
- e. Budding.

Use of Specialised Vegetative Structure

Vegetative propagation can be achieved through the use of specialised vegetative structures other than seeds. These specialised structures include bulbs, corms, rhizomes, tubers, runners, off shoots.

a. Bulbs – These are shooting stems with thick fleshy leaves. Stem buds develop at the axils of leaf scale to form miniatures or small buds (bulblets) which are known as offsets when grown to full size e.g. onion. Onion bulb

b. Corms – These resemble bulbs but have no fleshy leaves. They are solid stems structure with node and internode e.g. cocoyam. Cocoyam corm

c. Rhizomes- These are horizontal cylindrical stems growing underground with node and internode and readily produce adventitious roots. They may be slender and elongated or thick and fleshy. Growth proceeds from the terminal bud or through lateral shoot. Propagation is by cutting the rhizomes into several pieces, each containing a vegetative bud e.g ginger. A Rhizome

d. Tubers – These are thick portions of underground stem e.g. Irish potatoes. Nodes and internodes are present and nodes are arranged spirally. Tubers

e. Runners – Aerial stems develop from the leaf axil at the base or crown of plants. Runners grow horizontally along the ground and form new plants at one of the nodes along the ground eg. Strawberry.

f. Off Shoots – In many plants lateral shoots develop from the stem which when rooted serve to reduplicate the plant. This has been referred in horticultural terminology as offsets, crown, division or slips e.g. banana, pineapple.

Plants or crops are usually reproduced through sexual or asexual means depending on the mode that is suitable for the crop to produce at maximum productivity. While some crops reproduce through one means others reproduce through both means. Those crops that reproduce through sexual means have the advantage of providing genetically variability for modification while those that reproduce through asexual means have the advantage of producing true-to-type plants

Asexual Propagation Methods (Vegetative)

This method involves the induction of a vegetative section or part of a living plant to form roots and subsequently, developing it into a whole new plant. Plant multiplication does not involve the seed cycle (exchange of genetic materials) and therefore, it is the best way to maintain some species as clones; individuals identical to the parent.

Types of Propagating Materials (Propagules)

Seed

‘Seed’ is the generative part of the plant used for propagation. A seed is a small immature plant (embryo) protected by a seed coat or testa, which is formed from the outer layers of the ovule after fertilisation. The seed is the basic unit of propagating many tropical crops, including yam and fruits (pawpaw, passion fruit). Even in crop species whose primary mode of propagation is by vegetative means (e.g. mango, avocado pear), seed sowing constitutes an important method of regenerating new plants. Seeds are sown in three different ways namely: by broadcasting, drilling/row-seeding, and pocket drilling/ sowing in holes.

Advantages of sexual propagation are:

Ease of transportation of propagating materials,

Less cost, skill and work to raise seedlings,

Ease of vegetative propagation in mature plants, hybrid vigour and associated benefits of development of new varieties and

Wider adaptation to varying environments.

The disadvantages of sexual propagation include:

Slow seedling growth,
Non-prototype off springs,
Problem of ensuring uniform produce quality since most seeds originate from cross pollination (by wind, insects),
Weaker seedlings and
Longer period to plant maturity than vegetatively-propagated crop species.

Budding/Bud Grafting

It is a process consisting of the engrafting of the bud (scion) of a plant into the stem (stock) of another plant of the same genus. Generally, it is very suitable for propagating deciduous fruit (Citrus spp.) and shade trees. Budding of improved materials on regenerated chupons is one of the new methods of rehabilitating cacao in Nigeria. In the most common T-budding pattern, the desired scion from a young, actively-growing shoot of a chosen crop variety is immediately slid into a T-shaped slit on the rootstock. The joined bud and rootstock are held by a winding of rubber band/ special tape/wrap which holds it until sealed, which prevents drying or contamination of grafted materials. Chip budding is used for budding species whose barks do not “slip” (when cut, the bark easily lifts in one uniform layer from the underlying wood) easily without tearing.

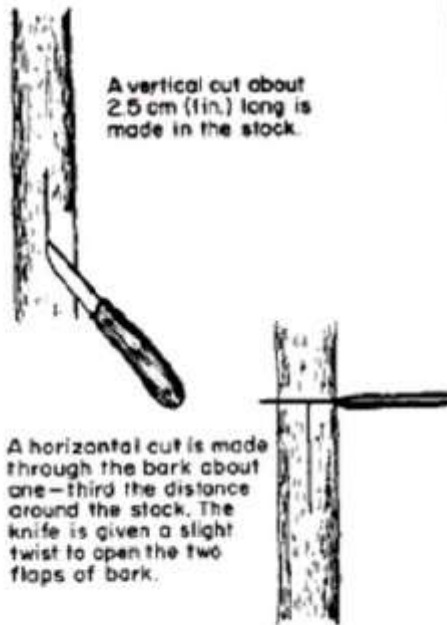
Advantages of bud grafting

1. Bud grafting is faster, easier and less messy than other forms of grafting discussed below.
2. Bud grafting allows the production of plants identical to a parent plant.
3. It may give increased productivity of crops through the hardness, superior rooting capacity, drought tolerance and insect or disease resistance of the rootstock.

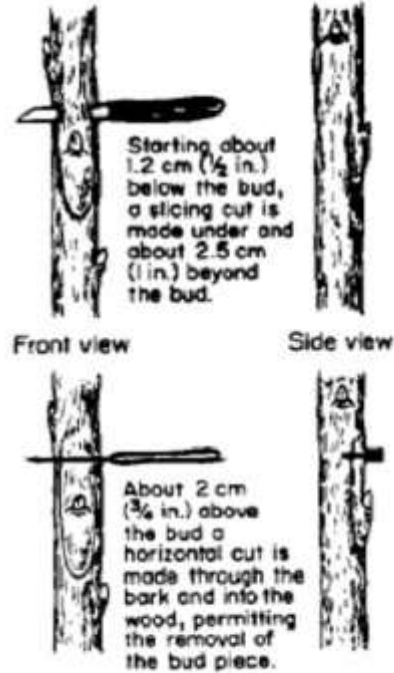
Disadvantages of bud grafting

1. The method is labour-intensive,
2. Requires great skill of nursery operations (and therefore, expensive) and
3. Can only be efficient when performed at very specific times when weather conditions and crop physiological growth status are optimum.
4. The vascular cambium of the both the bud scion and rootstock must be aligned to stimulate tissue growth on the basal ends before rooting.

PREPARING THE STOCK

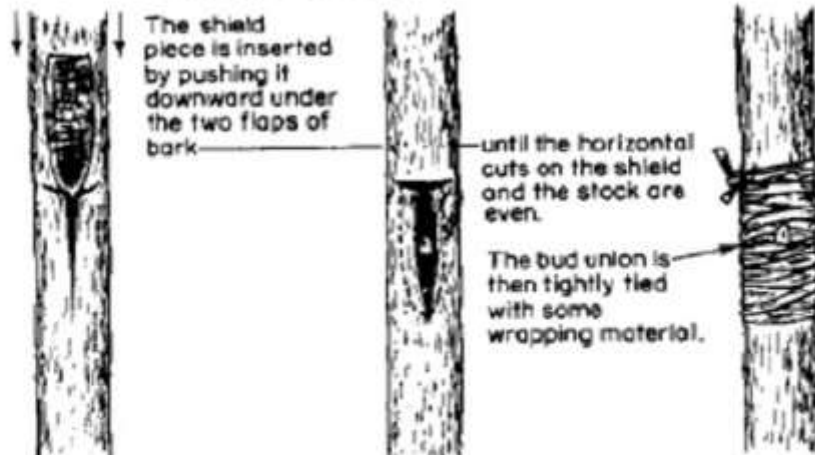


PREPARING THE BUD



INSERTING THE BUD INTO THE STOCK

INSERTING THE BUD INTO THE STOCK



Steps of T-budding and patch grafting

3.2.3 Grafting

GRAFTING

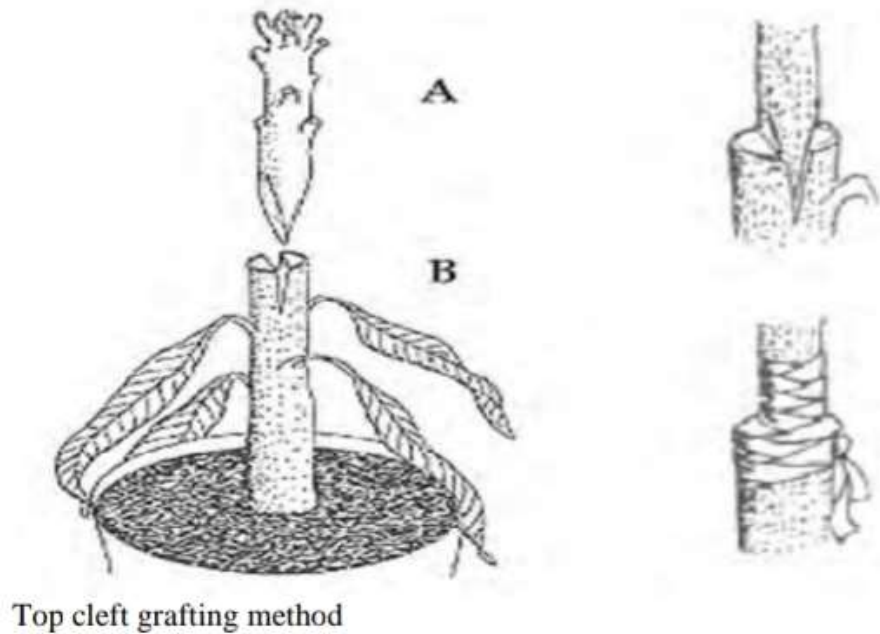
This is similar to budding in theory, but different in the sense that grafting involves the joining of the upper part (scion wood, 0.63-1.27cm diameter and only with leaf buds) of one plant to the under stock (rootstock) of another plant of the same species (clones, varieties). Grafting is an old art and science of plant propagation in pears, citrus, mangoes, grapes and other fruit trees, traceable back to 4000 years to ancient China and Mesopotamia. Some plants graft naturally, where two branches are in close contact over several years (e.g. ivy).

Advantages of grafting

1. Grafting allows gardeners to produce plants identical to a parent plant,
2. Allows growers to control size and shape of a tree or shrub (e.g. apples)
3. Gives more vigorous and earlier-fruiting plants.
4. Two varieties can be grown on the same tree to facilitate pollination (e.g. in apples).

Disadvantages of grafting

1. Grafting is labour-intensive, expensive, inefficient in poor weather and plant growth conditions,
2. Where cambiums of both scion wood and rootstock are not precisely aligned.
3. Graft incompatibility, sucker production in grafted plants and death of rootstock due to rooting of the scion arising from planting the graft union below the ground. There is a need to protect the grafted area from dislodging the scion out of alignment, especially by bracing.
4. There is a great risk of the top growth being very brittle thus, failing to harden off before cold weather.



Top cleft grafting method

Reasons for budding and grafting

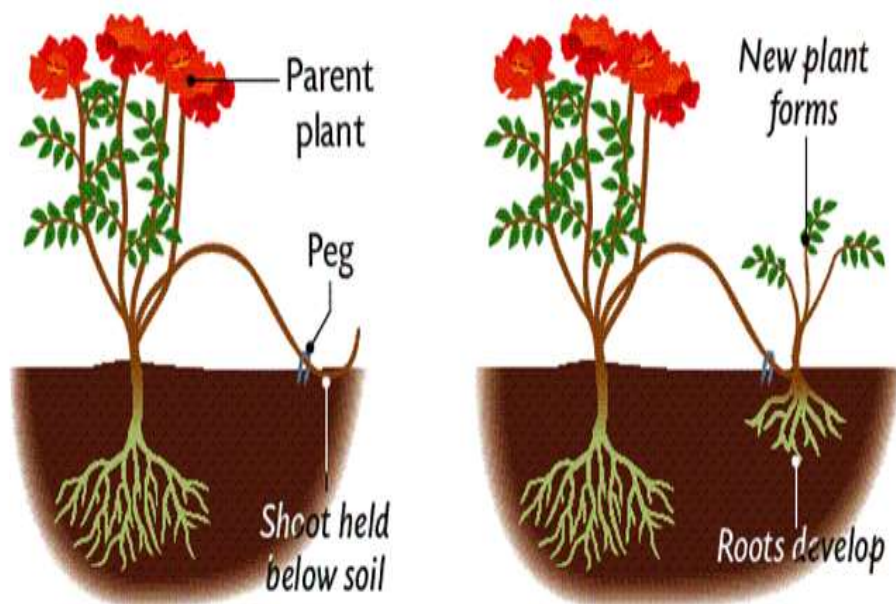
- i. Opportunity to change varieties or cultivars for crop improvement;
- ii. Optimising cross-pollination, especially in fruit trees that are not self pollinating;
- iii. Advantage of particular (desired) rootstocks, especially in respect of superior growth habits, disease and insect resistance, and drought tolerance;
- iv. Benefit from interstocks, valuable in a situation of graft incompatibility;
- v. To perpetuate clones by grafting onto seedling rootstocks when clones of plant species (e.g. conifers) cannot be economically reproduced from vegetative cuttings due to low rooting percentage of cuttings
- vi. To produce certain plant forms e.g. weeping or cascading forms as in weeping hemlock (*Tsuga canadensis* Carr var. *pendula*);
- vii. To repair damaged plants, arising from maintenance equipment, disease, rodents or winter storms, through in arching, approach grafting, or bridge grafting;
- viii. To increase growth rate of seedlings, especially in seedling progeny of many trees requiring 8-12 years to fruit with natural development; and
- ix. To facilitate virus indexing, through confirmation of presence or absence of the virus by grafting scions from the plant onto another plant that is highly susceptible and would quickly show symptoms of infection.

There are some rules, which must be taken consideration for any grafting method to be successful:

- Two incompatible plants cannot be grafted
- The cambium layers of the rootstock and the scion must touch
- The scion must be the right way up when you graft it
- You can grafting in any time of the year, but the best time for deciduous plants is, when the plant drops its leaves and is dormant
- Cool, cloudy day without wind prevents the graft from drying out, therefore this type of weather is the best for grafting
- The care activities are very important until the rootstock and scion are properly joined

Layering

This involves bending a branch/part of the stem of a growing plant and anchoring (with a rock or peg) and burying a portion of it, with a view to establishing a new root system at the point of contact between the bent part and the earth (i.e. on the shoots that are still attached to the parent plant). A light soil increases rooting success as will wounding or girdling of the buried portion. Treatment with a rooting hormone (e.g. Rootone, HormondinR, HormonexR) is most desirable, particularly one containing a fungicide. Plants with flexible branches are particularly suited to this method. As soon as the new plant is established, the connection with the parent plant is severed and the new plant becomes independent. Layering is a good propagation choice when only a few plants are needed. A heavy soil will reduce rooting success while covering the tip of the parent plant (bent shoot) kills it. Other types of layering are serpentine/compound layering, continuous/trench layering and mound/stool layering.



Layering of plants

Banking Up

This is the most common method to propagate pear, quince and apple rootstock (M type clones). This technique needs some preparative work. For about 2-3 years, we cut back the mother plant up to the surface level (or close to it), which will then result in a thicker root neck. This thick root neck will grow a number of stems. We then have to bank up the plant to 10-15 cm high (when the stems have an average 20-25 cm length). We can subsequently harvest the rooted stems at the end of the growing season. **Simple Layering**

This technique is commonly used for hazel-nut propagation. During the dormant season, stems are bent down into a 20-25 cm deep trench and covered with soil. The top parts of the stems, which usually have 2-3 buds on them, remain above the surface.

Radial or Chinese Layering

In this case, the whole stem is bent down into a 10 cm deep trench and covered with soil. We then have to bank up the suckers to 2/3 of their height on regular bases. This occurs when the sucker grows 10cm above the surface leaving only the top 1/3 of the plant free. This method is used mostly to propagate Gooseberry, Currant and Hazel-nut.

Air Layering

This method is used on the tip of the branch, when stems are usually younger than one year old. A strip of bark is cut approximately 2 cm wide on the stem about 20 cm from the tip (just below a leaf stalk, or joint). Once cut, a rooting hormone is applied and rooting material is placed under the strip. Finally, the cut is covered with a thin plastic bag, which is opened at both ends. Rooting material must be placed in the bag before it is sealed. This process must be completed during rainy conditions, when the air humidity is highest. Litchi, guava, macadamia and mango are propagated with this method.



Air layering

Cuttings

A cutting is a vegetative plant part which is severed from the parent plant in order to regenerate the parent plant (by regaining lost tissues), thereby forming a new plant. A cutting is therefore referred to as any cut portion of a plant which when placed in a suitable environment can regenerate into an independent plant. Both woody and herbaceous plants are asexually propagated by

cuttings of stems, leaves and roots. Cuttings regenerate new plants through change of mature cells into meristematic cells that are found at rapid growth sites like buds. As in layering, the use of rooting hormone such as auxins, cytokinins as a dip, preferably one containing a fungicide, helps to hasten rooting, increase number of roots, or gives uniform rooting, except on soft fleshy stems. Rooting medium may be coarse sand, vermiculite, soil, water or a mixture of peat and perlite. A major advantage of cuttings is the practical regeneration of whole plants from actively growing plant parts/organs. Cutting technique also, typically as with other asexual methods of plant propagation, produces several whole new plants, and which are genetically identical clones of the parent plant. Important disadvantages are that cuttings should be made as soon as possible after collection of plant material; not all species of plants can be propagated from cuttings (e.g. *Acacia* spp.); cuttings must be shielded from direct sunlight, especially if they are under glass or plastic; stock plants (plants used for asexual propagation) should be healthy and well-branched as should the tools and conditions for preparing cuttings to ensure healthy new clones; choice of correct rooting medium to achieve optimum rooting within the shortest possible time. There are many types of cuttings based on the vegetative part of the plant providing the cutting material. There are

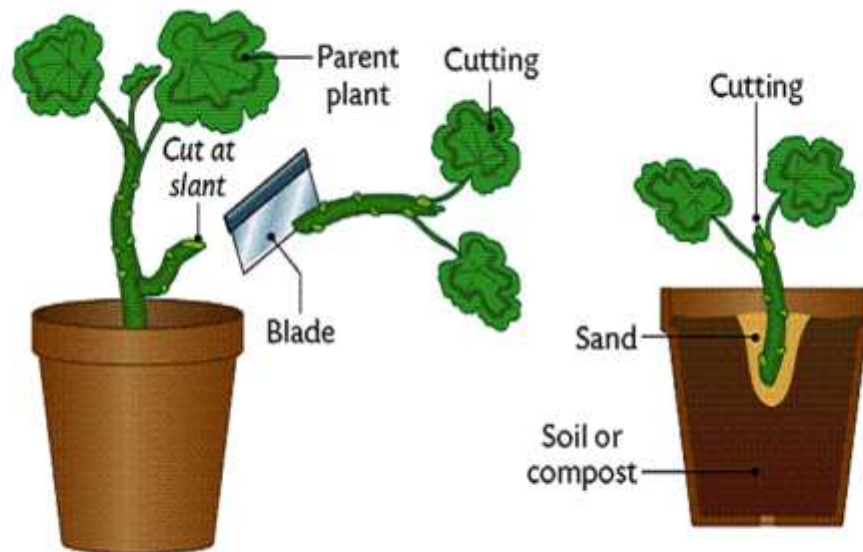
Stem Cuttings

This technique is the most commonly used method of propagating many woody ornamental plants and house plants e.g sweet potato, sugar-cane and cassava are food crops propagated from stem cuttings. Stem cuttings of many favourite shrubs are quite easy to root whereas those of a tree species are more difficult to root. A glasshouse is not necessary for successful propagation by stem cuttings but it is critical to maintain high humidity around the cutting. Facilities for rooting cuttings include flower pots, trays, small hoop frame and/or an intermittent mist system. Materials for making stem cuttings should be vigorous, new growth with no flower buds and free of diseases and insects. Cuttings should be 5.08-10.16 cm long, cut from older stems and have 2-3 leaves (2-3 nodes) attached. Dipping the base of the stem, including the node area, into a rooting powder stimulates rooting. The stem should however, be dry when dipped. Four main types of stem cuttings are identifiable based on the growth stage of the stock plant, which is very critical in the rooting of cuttings, namely:

- i. Herbaceous Cuttings- Cuttings taken from non-woody plants, such as perennials and house plants e.g. Chrysanthemums, rose. Cuttings are 5.08-15.24 cm stem pieces, with a terminal bud.
- ii. Softwood Cuttings- Cuttings taken from soft, succulent, new growth from non-woody stock plants, before the new growth starts to harden (mature). The cuttings are used to propagate flowering shrubs. They must be taken after rain or water is required to keep them cool in the morning. The larger diagonal cut gives more area to develop roots. Cuttings should be kept in water before rooting.
- iii. Semi-Hardwood Cuttings- Cuttings prepared from partially mature (firm) wood of the current season's growth, just after a flush of growth. The method is used for

propagating many broadleaf evergreen shrubs, some conifers, holly, rose and cacao (using the tips of shoots).

- iv. iii. Hardwood Cuttings- Cuttings taken from tissue that has become woody (firm) and the plant is mature and dormant with no signs of active growth. Several cuttings can be made from the same branch of some shrubs. Basal cuts should be just below a node, while the upper cut should be slightly above a bud. Cuttings should be kept moist until rooting. The system is most often used for deciduous shrubs and many evergreen species e.g. grape, fig and rose. The three types of hardwood cuttings are straight, mallet and heel cuttings.



Leaf Cuttings

Leaf cuttings are used almost exclusively for propagating a few indoor plants. Leaf cuttings do not include an auxiliary bud, and thus, can only be used for propagating plants that are capable of forming adventitious buds. The method involves the use of a healthy leaf blade or leaf without petiole in propagating new plants, following the same procedures as for stem cuttings, particularly treating leaf cuttings with growth hormones to stimulate rooting and quick bud development. There are several types of leaf cuttings, and for all of them, the old leaf is not part of the new plant and is thus, usually discarded. In most cases, the old leaf provides the energy food source for nurturing the newly generated plant e.g. *Bryophyllum pinnatum*.

- i. Whole Leaf with Petiole- This involves a whole leaf with about 3.81 cm of the petiole. The lower end of the petiole is dipped into a rooting medium (rooting medium may be coarse sand, vermiculite, soil water or a mixture of peat and perlite) after which one or more new plants form at the base of the petiole. The old petiole may be reused

after the new plants have formed their own roots. African violets and pepperoni are propagated in this way.

- ii. ii. Whole Leaf without Petiole- This method is used for propagated plants with sessile (petiole-less) thick, fleshy leaves. The leaf is inserted vertically into the rooting medium after which one or more new plants will form from the auxiliary bud. The leaf may be removed after the plant forms its own roots.
- iii. iii. Split-Vein- The veins on the lower surface of a leaf from the stock plant (e.g. Begonia and snake plant) are slit before the leaf cutting is laid on the medium. The rooting medium is used to hold down the margins of a curling leaf. A variation of this method involves inserting leaf wedges cut with at least one main vein into the medium with the main vein partially covered. In both cases, new plants are formed from the base of the split vein and leaf wedge.
- iv. iv. Leaf-Bud Cuttings- These are used for many trailing vines and when space or cutting material is limited. Each node on a stem can be treated as a cutting. A leaf-bud cutting consists of a leafblade, petiole, and a short piece of stem with an attached axillary bud. The cuttings are placed in the rooting medium with the bud covered (1.27-2.54 cm) and the leaf exposed. Rubber plant, Camellia, Rhododendron and blackberry are propagated using this method.
- v. v. Flower Stalks- This follows the same technique as leaf cuttings. It involves plant propagation from a flower stalk, usually with large leaf ears devoid of flower buds. This method enhances chimera production in African violet.

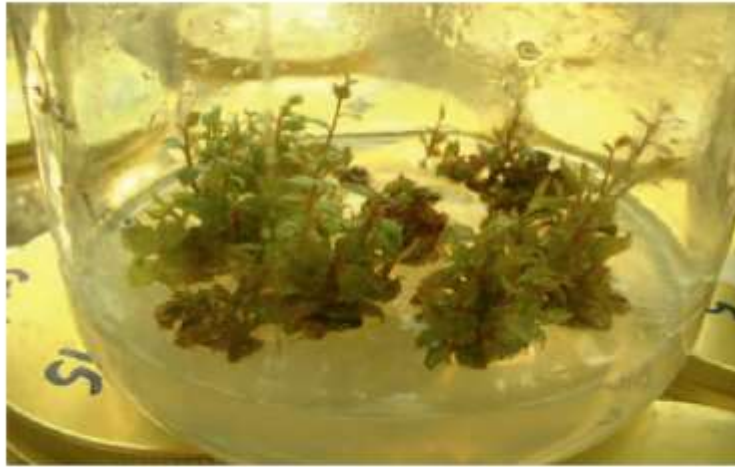
Root Cuttings

This involves the propagation of plants from a section of a root. In some species, the root cuttings produce new shoots which subsequently form their own root system whereas in others, root cuttings develop root system before producing new shoots. Plants propagated from root cuttings include blackberry and rose. In most cases, root cuttings of woody plants are usually taken during the dormant (inactive growing period) season when roots have large carbohydrate levels. Root cuttings can also be taken from actively-growing plants i.e. throughout the growing season. In plants with large roots that are normally propagated outdoors in a hotbed, the root cuttings should be 5.08-15.24 cm in length, with a straight cut at the proximal end and slanted cut at the distal end of the root cutting. In plants with small roots, the root cuttings are 2.54-5.08 cm in length and are laid horizontally about 1.27 cm below the soil or sand in a flat. The flat is then placed under shade, which is removed after new shoots appear.

Micropropagation or Tissue Culture

Tissue culture is a method used for vegetative propagation based on the phenomenon that any part of a plant from a single cell to a whole apical meristem can grow into a whole plant. The explant, the piece of the plant taken, is grown in a sterile artificial medium that supplies all

vitamins, mineral and organic nutrients. The medium and explant are enclosed in a sterile jar or tube and subjected to precisely control environmental conditions. In this method, individual or small group of plant cells (tiny pieces of bud leaf and stem) are manipulated in a way to enable them produce a new plant. Mass propagation of sugarcane, sweet potato tubers is achieved by in vitro culture of nodal segments in medium containing 9% sucrose under continuous darkness using Jar Fermentor Technique. Begonia and roses are also propagated by tissue culture using the meristem-tip.



Tissue Culture

Tissue Culture

The advantages of this method are:

speed and efficiency of plant propagation and
Production of disease-free (aseptic) plants.

Disadvantages include:

Spontaneous natural mutations and

Very exacting conditions for growing tissue culture materials, such as absolute sterile conditions, strict control of temperature, light, humidity and atmosphere with costly electronic sensors and computer equipment.

Plant or crops are usually reproduced through sexual or asexual means depending on the mode that is suitable for the crop to produce at maximum productivity. While some crops reproduce through one means others reproduce through both means. Those crops that reproduce through sexual means have the advantage of providing genetical variability for modification while those that reproduce through asexual means have the advantage of producing true-to-type plants.

VEGETABLE TYPE

1. Local or Indigenous Vegetables

These are native or traditional plants grown and eaten in a specific region. They grow well in local weather and soil, and people use them daily in traditional meals.

Local vegetables of economic importance in Nigeria

- i. **Fluted Pumpkin (*Telfairia occidentalis* - Ugu):** This is a vital economic crop across southern Nigeria valued for leaves and seeds.

Symptoms of pests and diseases:

- a. **Leaf Spot (Fungal):** Small yellow-brown or grey spots with dark borders on leaves that expand, merge, and cause the leaves to dry up and drop.
- b. **Anthraxnose (Fungal):** Leaves and vines shrink, lose color, and gradually wilt
- c. **Leaf-Eating Beetles and Caterpillars (Pests):** Irregular holes chewed through the soft green leaves, leaving skeletonized veins.

- ii. **African Spinach (*Amaranthus* spp. - Tete/Soko):** This is a highly lucrative short-duration leafy green. Improved cultivars feature heat tolerance, delayed bolting, and uniform dark-green or red-tinted high leaf mass.

Symptoms of pests and diseases

- a. **Aphids and Coreid Bugs (*Cletus* / *Aspavia* species) (Pests):** Clusters of tiny soft-bodied insects on stems and under leaves; leaves curl, wrinkle, and become sticky from honeydew secretions.
- b. **Whiteflies (Pests):** Small white insects that fly up when shaken; leaves show generalized yellowing (chlorosis) and weak growth
- c. **Wet Rot (*Choanephora cucurbitarum*):** This causes a rapid, water-soaked soft rot of stems and leaves. The infected soft tissue becomes covered with a characteristic dark, hairy, silk-like fungal growth with black tips, leading to quick collapse.
- d. **Damping-Off and Root Rot (*Pythium* or *Rhizoctonia*):** This affects seedlings, causing water-soaked or dark lesions at the soil line, resulting in sudden wilting and toppling over of young plants

- iii. **Jute Mallow (*Corchorus olitorius* - Ewedu):** This is an essential slimy soup vegetable. Improved lines offer high mucilage content, faster regeneration after cutting, and resistance to leaf-spot pathogens.

Symptoms of pests and diseases

- a. **Root-Knot Nematodes (*Meloidogyne* species):** Stunted plant growth, yellowing leaves, and wilting during hot afternoons; roots show swollen lumps or knots.
- b. **Variegated Grasshopper (*Zonocerus variegatus*):** Large ragged edges on leaves, severe defoliation, and complete stripping of young shoots.
- c. **Leafworm / Hairy Caterpillar (*Acraea terpsicore*):** Small to large circular holes or heavy skeletonization of leaves, leaving only the tough main veins behind

- d. **Anthracnose (*Colletotrichum* species):** Yellowish-brown, water-soaked, depressed spots on stems and leaves that turn dark brown or black, sometimes causing stem girdling and wilting.
- iv. **Waterleaf (*Talinum triangulare*):** This is a fast-growing wet-season and irrigated crop. Selected local morphotypes provide succulent broad leaves and higher iron content.
- v. **Tomato (*Solanum lycopersicum*):** Improved varieties bred by research institutes include *Roma VF*, *UC 82B*, and *Tropicana*, prized for uniform ripening, extended shelf life, heat-setting capacity, and resistance to bacterial (*Fusarium* and *Verticillium*) wilts.
- vi. **Pepper (*Capsicum* spp. - *Rodo/Tatase*):** Improved lines like *NHV-1* and high-yielding *Scotch Bonnet* hybrids offer resistance to mosaic viruses and heavy fruit load
- vii. **Okra (*Abelmoschus esculentus*):** Improved varieties such as *LD88* and *NHokra 47-4* provide early maturity, high mucilage, and resistance to yellow vein mosaic virus.

Symptoms of pests and diseases

- a. **Yellow Vein Mosaic Virus:** Veins of the leaves turn completely yellow, creating a net-like yellow pattern while green patches remain, severely stunting fruit production.
- b. **Cercospora leaf spots / Cercospora leaf blight / Black mould:** The disease shows on the lower surface of the leaves as greyish spores that later become black and sooty.
- c. **Cotton strainers:** Juvenile fruits dry out and drop off while the pods near maturity stage develop spots and dry out.
- d. **Fusarium wilt:** In the event of a severe attack, many leaves droop and become chlorotic leading to overall wilting of the plant which subsequently dries out and dies.

Growth requirements for local vegetables

1. **Soil Type:** Local vegetables thrive well in loose, well-drained loamy or sandy-loam soils or in native, diverse soils, including slightly degraded or heavier tropical soils, provided they have basic organic nutrients.
2. **Water availability:** They are highly resilient; many tolerate irregular rainfall or brief dry spells once established.
3. **Climate:** They are adapted to warm, tropical, or subtropical temperatures and local daylight durations.
4. **Topography:** They require flat to gently sloping land (ideally under 1.5% to 2% grade) to prevent severe soil erosion, ensure proper surface drainage, and ease mechanized or manual field operations.
5. **Accessibility:** Vegetable establishments require dependable all-weather rural roads for quick transport of fragile produce, seeds, and fertilizers to and from the farm.
6. **Pests and diseases:** Local vegetables show higher natural resistance to endemic local pests and fungal strains.
7. **Market:** They are largely sold in local, traditional, and rural community markets with steady, everyday high volume demand.

Cultural and maintenance practices for growing local vegetables

Growing local vegetables in Nigeria such as fluted pumpkin (*ugu*), jute mallow (*ewedu*), African spinach (*tété/soko*), and bitter leaf, involves distinct seed selection, raised bed preparation, organic manure enrichment, and manual weed and water management tailored to the tropical wet and dry seasons.

1. **Land preparation: Raised beds and ridges should be built** to ensure proper soil drainage and prevent root rot during heavy tropical rains.
2. **Planting: For seasonal timing**, planting should be done in rainy season (April–October) or supported via irrigation during the dry season (November–March). As for the sowing methods, direct seeding for robust seeds like *ugu* (fluted pumpkin), and nursery sowing or broadcasting for tiny seeds like Spinach and jute mallow.
3. **Soil and fertility maintenance:** The soil is often frequently fertilized using household organic refuse or well-composted farmyard manure rather than costly chemical inputs
4. **Mulching:** A mulch is a layer of plant residue or other materials which is applied to the surface of the soil in order to reduce evaporation, run-off or to prevent weed growth. The purpose of mulching is to conserve soil moisture. Mulching also ensures clean fruit, hasten maturity and increase yields. Most vegetables grown in the humid or semi-arid region, particularly those with fairly long duration such as pepper, egg plant and tomatoes, benefit from mulching.
5. **Weed Control:** Regular manual weeding or hoeing should be done to stop weeds from competing for essential nutrients and moisture. Weeds can reduce yield up to 40-60% if not controlled. Weeds can be controlled using cultural, physical, chemical and biological methods. Weed seeds and rhizomes can be killed using physical method during land preparation by burning. Mulching of soil can also be used to smother weeds. Intercropping of spreading vegetables like melon with some erect and broad leaf crop like okra can provide ecological protection against weed development. Hoeing, pulling and roguing are carried out during the early stages of growth. Chemical weed control is applied in commercially grown vegetable crops.
6. **Water Management:** Frequent early-morning or evening watering via cans, hoses, or localized irrigation is required to maintain turgor pressure without over-saturating the soil.
7. **Pest and Disease Control:** Field sanitation and organic deterrents like neem oil can be used to manage local insect pests safely

2. Exotic Vegetables

Exotic vegetables are uncommon or non-native produce types valued for unique flavors and appearances, featuring popular choices like Romanesco, bok choy, and globe artichokes. These

specialized crops are frequently imported or grown in controlled environments to supply upscale markets and modern culinary recipes. Common exotic vegetables in Nigeria include:

I. Cabbage (*Brassica oleracea*)

Improved Varieties: Copenhagen Market and Golden Acre (valued for tight head formation and adaptability).

Symptoms of pests and diseases

- a. **Black Rot:** V-shaped yellow lesions starting at the leaf edges, blackening of veins.
- b. **Downy Mildew:** Yellow patches appear on the upper leaf surface with a white or gray powdery mold underneath.
- c. **Cabbage Worms:** Velvet-green caterpillars eat large irregular holes in the outer leaves and leave dark green droppings behind.
- d. **Cabbage Aphids:** Tiny clusters of green or gray bugs gather on the undersides of leaves, causing leaves to curl, turn yellow, and look stunted

II. Carrots (*Daucus carota*)

Improved Varieties: Nantes and Chantenay (selected for uniform root length, sweetness, and crisp texture)

Symptoms of pests and diseases

- a. **Alternaria Leaf Blight:** Greenish-brown, water-soaked lesions which enlarge, turn brown to black, and often develop a yellow halo. Older leaves turn yellow, collapse, and die.
- b. **Cercospora Leaf Spot:** Small necrotic flecks on younger leaves that develop a yellow halo, causing leaves to curl and wither
- c. **Root-Knot Nematodes:** Galls on roots, forked and deformed carrots, and stunted growth. Plants may also show leaf yellowing.
- d. **Aphids:** Curled, distorted leaves and stunted plant growth due to sap-sucking.

III. Lettuce (*Lactuca sativa*)

Improved Varieties: Great Lakes and Iceberg (bred for slower bolting in warmer weather).

Symptoms of pests and diseases

- a. **Bacterial Soft Rot:** Water-soaked, slimy, and soft rotting lesions develop on lower leaves and stems touching the ground.
- b. **Downy Mildew:** Chlorotic spots on the upper surface of the leaves and a whitish, floury powder on the underside of the leaves
- c. **Bacterial Leaf Spot:** Small, angular, water-soaked spots appear on outer leaves which later turn brown, dry out, and collapse, leaving dead tissue.
- d. **Aphids:** Curled, distorted leaves and stunted plant growth due to sap-sucking. Plant can turn yellow or wilt. Sticky honeydew secretion builds up, attracting ants or growing black sooty mold.

IV. Sweet Pepper (*Capsicum annuum*)

Improved Varieties: HortiPep1 and HortiPep2 (yellow aromatic types bred locally for disease resistance).

Cultural and maintenance practices for exotic vegetables

Growing exotic vegetables like broccoli, lettuce, and bell peppers in Nigeria requires protected greenhouse structures, precise drip irrigation, and specialized soil or soilless management to counter the tropical lowland heat

1. **Site and Climate Adaptation:** Traditionally grown in cooler high-altitude regions like the Jos Plateau, exotic vegetables are now cultivated in lowland urban areas using net-houses or greenhouses to manage intense sunlight and heat.
2. **Land preparation:** It is best to use well-tilled sandy loam soil shaped into raised beds or ridges to ensure optimal root penetration and prevent waterlogging.
3. **Nursery establishment:** High-value hybrid seeds are first raised in seedling trays under shade to protect young plants from heavy tropical downpours and intense solar radiation before field transplanting
4. **Water Management:** Controlled water delivery via drip irrigation systems maintains uniform soil moisture and prevents root stress or fungal splashing. Young vegetable seedlings in the nursery or in the field should be watered in the early morning or in the evening. Watering should be done before transplanting particularly in the evening. Overwatering can be very harmful and can encourage the development of pathogenic diseases and also cause mechanical damage to the seedlings.
5. **Nutrient Application:** Balanced fertilization combines organic compost with targeted chemical fertilizers or liquid foliar micronutrient sprays to maximize yield
6. **Pest and Disease Control:** Regular field scouting and biological controls, such as beneficial microbes like *Trichoderma* or organic deterrents like neem oil, can manage common pests like thrips and spider mites.

Growth requirements for exotic vegetables

1. **Soil Type:** They require deep, loose, well-draining loam or sandy loam rich in organic matter, with an optimal pH range of 6.0 to 7.5.
2. **Water Availability:** They demand precise, consistent watering schedules and/or regular irrigation (such as drip systems) without waterlogging. Drought stress or erratic rainfall causes bolting, bitterness, or poor root/head formation.
3. **Climate:** Originating mostly from temperate zones, they prefer cooler temperatures (15°C to 24°C) or must be grown in high-altitude tropical uplands or climate-controlled greenhouses.
4. **Topography and Accessibility:** Similar to the local vegetable requirements.
5. **Disease and Pest:** They are highly sensitive to non-native or local tropical pathogens, and they therefore require strict integrated pest management (IPM) and protective netting or greenhouses.

6. **Market:** They are often targeted at high-end urban supermarkets, hotels, restaurants, and export channels, fetching premium prices but requiring cold-chain storage

Other cultural and maintenance practices for growing vegetables

1. **Thinning:** Thinning of vegetable is done to reduce the number of seedlings per stand when planted in situ. It is also used to obtain regular spacing and plant population when sowing is done in drills by removing the weakest seedlings or those infected by diseases or damaged by insect pests.
2. **Supplying:** This is the practice of providing missing stands of vegetables planted by direct sowing as a result of poor emergence or when seedlings are damaged by pests. The essence of seed supply is to maintain correct plant population. Supplying of seeds has to be carried out as early as possible after emergence.
3. **Staking:** This is usually required for vegetables with climbing growth habit such as fluted pumpkin, ito-melon or those with weak stems such as tomato. Staking is done to provide support for the plant to climb and display the leaves for photosynthesis to take place.. The support allows the plant to carry more load without touching the soil thus enhancing the quality of the fruit.. Stake can be made from bamboo or other available wood.

IMPORTANCE OF VEGETABLES IN HUMAN DIET

Vegetables supply most of the nutrients that are deficient in other food materials. This includes supply of minerals, especially calcium and iron. Vegetables are acid neutralizers e.g. okra, *Corchorus* spp neutralizes the acid produced from the some fruits. Vegetables prevent constipation and promote digestion as a result of fibres/roughages obtained from okra, cucumber, amaranthus, lettuce and cabbage. Vegetables are rich sources of vitamins A, B, and C which helps to lower susceptibility to infection. e.g.: Carrots, sweet corn, amaranthus and celosia provide Vitamin A; Bitter leaf, water leaf, *solanum* and celosia provide Vitamin B; Tomatoes, carrots, lettuce, cabbage and amaranthus provide Vitamin C.

Also, some vegetables are rich sources of carbohydrate e.g. potatoes, sweet corn, carrot etc. Green beans and peas are cheap sources of protein. Vernonia (Bitter leaf), Amaranthus and *Telfeira* provide some amount of protein in human diet.

Vegetables are generally needed to have balanced diets and overcome nutritional deficiencies. Vegetables make our staple food more palatable and enhance their intake.

The practice of cultivating vegetables and spices for food, composite seasoning and medicine is referred to as Olericulture. Olericulture is one of the four general and sub-categories of horticulture after Pomology, Floriculture, and Landscape gardening/

VEGETABLE GROWING ENTERPRISES

Several vegetable growing enterprises have developed owing to rapid urbanization as well as socioeconomic and political situations. The enterprises are:

- 1. Home Gardening:** This is principal source of fresh local vegetable supplies for most homes. It supplies an important part of the family needs and it is also a source of additional tax-free income for the farmer.
- 2. Market Gardening:** This is a typically smaller, highly diversified local operations that sell directly to local consumers, farm stands, or restaurants.
- 3. Truck Gardening:** Sometimes used interchangeably with market gardening, it is characterized by larger-scale, less diversified operations that focus on mass-producing select crops to be transported to distant or regional distribution hub. The name does not originally come from transportation motor vehicles. Instead, it derives from the old French word *troquer*, which means "**to barter or exchange**"
- 4. Commercial Production:** This is the principal source of vegetables for big market. It is more extensive and specialized than market gardening. The area of production is determined by climatic and edaphic factors.
- 5. Production for Processing:** This is a principal source of fresh vegetable for processing industries. The scale of operation is similar to that of commercial production.
- 6. Vegetable Forcing:** This is the practice of producing vegetables out of their normal production season. It may be accomplished by modifying the growing environment such as heat provision or protection from cold, etc.
- 7. Controlled Environment Agriculture:** This involves the practice of modifying the natural environment for optimum plant growth and production of growth factors such as light, air, temperature, water, relative humidity, etc.
- 8. Vegetable Seed Production:** This is a specialized agricultural industry for seed production and processing. The enterprise only deals with seed production and not fresh vegetables for consumption.

IMPORTANCE OF DISEASE AND PEST CONTROL IN VEGETABLES

Attack by pest and disease is one of the major factors militating against increased vegetable production in the tropics. Insects and disease infestation in vegetable crops bring about heavy losses through yield reduction, lowered quality of produce, increased costs of production and harvests. Disease and pest control in vegetables is vital to enhance crop yield, improve market quality, and ensure food security. Managing these threats early protects plant health and prevents total crop failure. Control measures include:

- i. planting of high quality, disease-free seed
- ii. use of disease resistant varieties.
- iii. use of vegetables best suited to climate and soil.
- iv. weed control

v. destroy the remains of annual crop after harvest to prevent pest build-up.

Benefits of pests and diseases control in vegetable include:

- i. Stopping insects and fungi from destroying leaves, roots, and fruits so more food can be harvested would lead to higher yields.
- ii. **Better Quality:** Preventing spots, holes, and rot, makes vegetables look clean and safe to sell.
- iii. **Food Safety:** Pests and diseases control lowers the risk of plant diseases reaching consumers and keeps food supplies safe.
- iv. **Reduced production cost:** Effective control of pests and diseases in vegetable reduces the high cost of fixing ruined crops or buying extra chemicals later.

HARVESTING OF VEGETABLES

Harvesting is the separation of the portion of need from the mother plant. Most vegetables, particularly the fresh leafy and fruit types, are highly perishable. They quickly grow and as soon as they reach the peak maturity (i. e. right harvesting stage), they start to deteriorate at a fast rate. This normally depends on the environmental conditions – warm weather encourages quicker maturation and also deterioration

Harvesting of vegetables may be carried out once or over time or repeatedly. Leafy vegetables may be harvested by total removal or by ratooning. The seed and the fruit vegetables may be harvested once, or by topping or digging or lifting. Fruit vegetables may be harvested over a period of time by picking. Vegetables need timely harvest in order to obtain the best quality at harvest and thereafter. Too early or too late harvesting can result in poor quality and affect the shelf life of vegetables and their ability to be transported to distant markets. Depending on the type, harvesting can be done before physiological maturity provided that the vegetable has reached horticultural maturity

Different methods, known also as maturity indices (index), can be used by farmers to determine when to harvest different types of vegetables. These methods include:

- i. Changes in chemical composition (e. g. accumulation of sugars in fruit vegetables, change in acid content (sources), loss of bitterness, change in starch content, production of certain gases e. g. ethylene, etc.). These indices require suitable equipment for their proper determination
- ii. Changes in physical or visual appearance (e. g. weight and size changes, change in colour, disappearance of surface structures such as hairs, development of a glossy or shiny appearance, e. g. eggplant)
- iii. Development of an abscission point (a weak layer of tissue called the abscission layer develops at the base of each leaf stalk (or petiole) or fruit stalk and allows the leaf or fruit to fall easily during harvesting or when they fall naturally from the parent plant)
- iv. Aroma development

Harvesting is preferably done in the early morning hours, preferably before the sun comes up, because at this time the vegetable leaves and/or fruits will be very firm and can store longer. This

high firmness is a result of greater moisture absorption than loss by the plants during the cool night environment. Harvesting late when it is hot and the vegetables are warm or hot may speed up wilting and reduce shelf life. Suitable harvesting tools and containers must be used to reduce any form of damage to the produce. Poor harvesting practices and timing can result in higher levels of bruising and other types of physical damage as well as wilting of the harvested produce. Knives, pruning shears and other sharp-edged tools can be used for harvesting different types of vegetables without much damage.

POST – HARVEST HANDLING OF VEGETABLES

These are activities carried out after harvesting till disposal. These activities include:

1. **Field and post field handling:** From the time when vegetables are harvested until the time that they are consumed they undergo different forms of handling and environments. Handling happens during the following processes: i) picking or harvesting, ii) cleaning, iii) grading and or sorting, iv) packing v) transfer to storage, vi) cooling, and vii) transporting from the field to packing places in shed houses, or from the farm to the markets. This intensive handling can result in damage such as bruising, cuts, cracking, and compression. Other forms of injury, e.g. of a physiological nature, can also occur during handling, for example through exposure to ethylene gas which can cause damages to sensitive vegetables.
2. **Cooling or shading:** All harvested produce must be kept under well ventilated shade to cool after harvest. Techniques that can be employed include hydrocooling for leafy vegetables, forced-air cooling for packed snap beans, precooling through vacuum cooling and ice.
3. **Curing:** Vegetables like onions and potatoes require curing before treatment. Potatoes intended for long term storage (up to 12 months) should be first stored at around 15 to 20 °C and 90 to 95 % relative humidity for about 10 days to cure them before storage. Curing stimulates wound healing on the potato skins and helps the skins to thicken thereby reducing water loss from the tubers during storage. These wounds may have been caused by damage during digging at harvest and bruising or cuts during handling after harvest. If not cured, the tubers will lose a lot of water through the wounds and result in economic weight loss. The wounds will also provide entry to pathogens and hasten rotting during storage.
4. **Cleaning:** Some vegetables should be cleaned before storage to remove, soil/dirt, dust, plant debris, insects, and other unwanted materials adhering to the vegetables. In most cases, the cleaning before storage should be dry, because wet treatments can help to spread bacterial or fungal pathogens on the vegetables resulting in rapid deterioration of the produce. After storage and shortly before marketing, a final cleaning step is made. If they have to be washed, drinking quality water must be used. Dry vegetables (e.g. bulb onions) must not be washed. After cleaning, the produce is sorted/graded, inspected for damage and kept under cool conditions (at least under shade) in order to increase shelf life.
5. **Sorting and Packaging:** If no field sorting has been done, soon after cleaning the vegetables must be sorted or graded to achieve uniformity based on size, shape, weight or colour, etc. Sorting must be done regardless of whether the vegetables are destined for fresh or processed

market. Packages should contain the produce and offer them good protection from damage or injury, contamination and water loss. There are many different types of packages available. They differ in design, size, type of material, and other characteristics. Bags, crates, hampers, baskets, cartons, bulk bins, and palletized types are examples of packages. Packages must have smooth insides to avoid inflicting scratching damage on the produce. The types of packages used must make the vegetables convenient for handling and distribution (including transportation and marketing). When packaged, the produce should fit well inside the container, with little wasted space. Cost should be considered in selecting the type of package to use. Environmental considerations must be taken into account also – biodegradable, renewable and recyclable types cause less harm to environment compared to durable plastics for instance. Finally, packing materials must not contain chemicals which contaminate the foods.

6. **Storage:** Due to their high perishability, most vegetables cannot store well without proper storage conditions and facilities. To reduce water loss and slow down deterioration in harvested vegetables, it is generally advisable that vegetables are stored at low or relatively low temperatures and high humidity. Optimal storage temperatures are higher for those vegetables that originated from the tropics and subtropics. Those from temperate climates require lower storage temperatures. The specific requirements differ from crop to crop. Examples of farm level storage strategies for vegetables are pot-in-pot evaporative cooling, underground storage pits and cooling buildings. Refrigerator can also be used to store some vegetables but freezer is not considered appropriate.
7. **Processing:** Once harvested, vegetables start to spoil or lose their quality unless control measures are taken to slow down the process. The loss of quality can be due to natural senescence (ageing processes) or induced by physical, chemical, enzymatic, or microbial processes. Insects and other pests such as rodents can cause and or accelerate spoilage in harvested vegetables. For example, wilting or other forms of dehydration may render the produce look unattractive. For seasonal vegetables, preservation is also important to extent their availability beyond the normal period.

Examples of processing/ preservation techniques at farm level

Some of the common farm level preservation methods for vegetables include:

- a. Boiling/blanching: This process arrests enzymatic reactions in the produce and prevent or reduce several physiological processes which can lead to rapid deterioration.
- b. Dehydration (with and without salt, with and without grinding into powder): This can be done under the sun, in solar dryers or electric powered dryers, or using fire depending on the product. If drying in the open sun, the produce should be protected from contamination by dust and other impurities.
- c. Purees and chutneys for e. g. tomatoes (and bottling into sealable jars)
- d. Fermentation (results in formation of acids which then prevent certain microorganisms from growing, hence protect the product)
- e. Pickling of onions and cucumbers in a salt solution or a vinegar solution

- f. Using vinegar
8. **Transportation:** Suitable means of transportation that would ensure that the vegetables retain their quality should be used.
9. **All activities which happen during the marketing stages until the vegetables are consumed:** This is the movement of the vegetable from the farm gate to an accessible area for the consumers.

ORNAMENTAL PLANTS AND FLOWERS

Ornamental plants are defined as plants cultivated primarily for aesthetic and decorative purposes, valued for their attractive flowers, foliage, and fragrances, rather than for food production. They are commonly used in landscape design to enhance the beauty of environments such as gardens and public spaces. Examples include **Hibiscus** (*Hibiscus rosa-sinensis*), **Bougainvillea** (*Bougainvillea spectabilis*), **Ixora** (*Ixora coccinea*), **Allamanda** (*Allamanda cathartica*), **Pride of Barbados** (*Caesalpinia pulcherrima*), **African Blood Lily** (*Scadoxus multiflorus*), **Marigold** (*Tagetes erecta*), **Dumb Cane** (*Dieffenbachia seguine*), **Areca Palm** (*Dypsis lutescens*), **Mother-in-Law's Tongue** (*Sansevieria trifasciata*) and **Peace Lily** (*Spathiphyllum wallisii*).

Classification of ornamental plants

Ornamental crops can be grouped based on their lifecycle and functional roles.

Classification by Life Cycle

1. **Annuals:** These complete their entire life cycle in one growing season or a year. Examples are marigolds and zinnias.
2. **Biennials:** These take two years to complete their life cycle, growing foliage the first year and flowering the second. Foxgloves is an example of such crops.
3. **Perennials:** These crops live for more than two years, returning each season from roots or woody stems. Examples include hostas and peonies.

Classification by Functional Roles

1. **Ground Covers:** These are low-growing plants that spread to cover the soil and prevent weeds e.g., creeping juniper.
2. **Climbers / Vines:** These are plants with climbing stems or tendrils used for vertical coverage on walls or trellises e.g., ivy, clematis.
3. **Accent / Specimen Plants:** These are unique or striking plants used to draw the eye and create a focal point in a garden e.g., Japanese maple.
4. **Hedge / Border Plants:** These are dense shrubs or flowers planted in rows to mark boundaries or line pathways e.g., boxwood.

NURSERY

A nursery is a place where young plants are raised under intensive care before transplanting into the field.

Types of Nursery in Ornamental Horticulture

Classification by Function and Market

1. **Retail Nurseries:** These sell direct-to-consumer for home gardens and small landscaping projects.
2. **Wholesale Nurseries:** These cultivate large volumes of plants for commercial landscapers, city projects, and retail garden centers.
3. **Propagation/Mother Block Nurseries:** This nursery types focus on generating specialized rootstocks, cuttings, and high-value hybrid varieties.

Classification by Plant Type Produced

1. **Floriculture/Annual Nurseries:** These type of nursery specializes in seasonal flowering bedding plants like marigolds and petunias.
2. **Woody Ornamental Nurseries:** These nurseries mainly grow landscape trees, perennial shrubs, and hedge plants.
3. **Indoor/Houseplant Nurseries:** Shade-loving foliage plants, ferns, and succulents suited for indoor spaces are cultivated in this nursery type.
4. **Specialty Nurseries:** These focus on unique collections like bonsai, cacti, or palm

Classification by Structure and Duration

1. **Temporary Nurseries:** These are small-scale, short-term setups built for a specific seasonal demand or local landscape project.
2. **Permanent Nurseries:** These are long-term commercial operations equipped with permanent mother blocks and storage structures.
3. **Open Field Nurseries:** These are nurseries in which plants are grown outdoors in natural soil beds with minimal protection.
4. **Protected/Hi-Tech Nurseries:** These utilize greenhouses, shade-net houses, and poly-tunnels to control temperature and light for delicate ornamental species.

Technologies of nursery practices in ornamental horticulture

Establishing an ornamental plant nursery requires careful site selection, proper soil and water preparation, and effective propagation methods. The key steps involve choosing a level, well-

drained location near a reliable water source, preparing fertile growing media, and utilizing correct propagation techniques.

1. Site Selection and Layout

- a. **Terrain and Soil**: A flat or gently sloping area (2–5% slope) that does not flood or stay waterlogged should be chosen. Ensure the soil is loose, well-draining soil (sandy loam) with a balanced pH between 5.5 and 7.0, free of excess salts.
- b. **Water Supply**: The nursery area should be nearer to reliable, year-round water sources.
- c. **Shade and Protection**: Set up partial natural or artificial shade netting to protect tender young flowers and seedlings from harsh sunlight and heavy rain. Install fencing to keep out stray animals.
- d. **Accessibility**: It must connect well to major roads for easy transport of plants, soil mixes, and equipment using vehicles.
- e. **Sunlight and Air**: The site should receive adequate sunlight while offering space for shade structures or windbreaks.
- f. **Labour**: There should be proximity to a workforce with both skilled and unskilled gardening knowledge.
- g. **Security**: There should be space for proper fencing to protect valuable ornamental stock from thieves and stray animals.

2. Soil Preparation and Potting Media

- a. **Clearing**: Clear the land completely of weeds, large stones, and wood debris using basic tools like hoes and machetes.
- b. **Media Mix**: Prepare a rich, well-draining potting mixture using fertile loamy soil blended with organic compost or farm manure.
- c. **Containers**: Fill nursery beds, polybags, or clay pots with the prepared soil mixture to give young roots a clean and safe environment to start growin

3. Plant Propagation Techniques

- a. **Sexual Propagation (Seeds)**: Sow clean seeds directly into specialized seedbeds or germination trays, keeping them moist until sprouts appear.
- b. **Asexual Propagation (Vegetative)**: Multiply many perennial and shrub varieties using stem cuttings, root cuttings, leaves, or underground structures like bulbs, corms, and rhizomes.

4. Daily Management and Care

- a. **Watering**: Water the young plants regularly, often twice a day during dry spells (in the early morning or in the evening), ensuring root balls do not dry out completely.
- b. **Weeding and Spacing**: Pull out unwanted weeds frequently so they do not steal food and water from your ornamental crops. Arrange pots with enough space and light so plants grow straight and healthy

Layout of buildings in the nursery

The best layout for an ornamental horticulture nursery follows a logical workflow from propagation to sales, grouping key zones like the office, propagation houses, shade houses, and storage areas to save labour.

- 1. Administrative and Sales Area:** Place the office and sales counter near the main entrance for good visibility, customer welcome, and easy business transactions.
- 2. Propagation Zone:** Position mist chambers, germination houses, and propagation structures in a quiet, protected rear or side section close to a reliable water source.
- 3. Production & Growing Structures:** Set up greenhouses, lath houses, and shade houses in continuous rows with a 10-foot minimum space between structures for air flow, movement, and drainage access.
- 4. Hardening-Off Area:** Locate transition and open growing beds directly adjacent to production houses to safely expose plants to outdoor conditions.
- 5. Service and Storage Buildings:** Put the tool sheds, fertilizer/soil stores, and packing sheds centrally or near service driveways so staff can load supplies without crossing customer retail paths.
- 6. Traffic Separation:** Keep retail buyers separate from heavy production workflows during busy seasons to prevent accidents and congestion.
- 7. Drainage Slope:** Build on a gentle 2% slope so surface water drains naturally without washing away soil or flooding building foundations.
- 8. Access Paths:** Design wide main aisles that accommodate wagons, carts, and small utility vehicles